High-dimensional portfolio optimization using GARCH-EVT, R-vine Copula, and SCAD penalty with weight constraints

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Abstract

This study presents a forecasting methodology for high-dimensional portfolios by integrating ARMA-GARCH models with Extreme Value Theory (EVT) to model asset return marginals and tail behavior. To address challenges from high dimensionality—such as complex dependence structures, more intricate trading rules, and higher transaction costs—our approach employs R-vine copulas to capture inter-asset dependence and incorporates the SCAD penalty to enhance sparsity and robustness. To address numerical solvability issues, we solve the constrained portfolio optimization problem using penalized quantile regression. In the empirical analysis, we first use ten sector indices to validate the effectiveness of the R-vine copula dependence structure. We then apply the full framework to high-dimensional portfolios constructed from constituents of the CSI 300 Index. Investment performance is evaluated along five dimensions: portfolio sparsity and stability; tail risk (VaR and CVaR); cumulative returns; risk-adjusted performance (RAROC and Sharpe ratio); and computational efficiency. Statistical differences in cumulative returns, RAROC, and Sharpe ratios across portfolios are examined using the Scheffé test. Empirical results demonstrate that the proposed framework effectively captures large-scale asset dependence, improves portfolio sparsity and stability, reduces tail risk, and enhances risk-adjusted returns.

Keywords: High-dimensional, Portfolio optimization, R-vine copula, SCAD penalty, Penalized quantile regression

1. Introduction

The rapid expansion of financial markets has led to an increasing dimensionality of investable assets, posing a significant challenge for portfolio management. In high-dimensional settings where the number of assets approaches or exceeds the available sample size, traditional portfolio strategies often struggle effectively to balance risk and return. Although Markowitz's mean-variance (MV) framework forms the basis of modern portfolio theory, its practical application is limited by a high sensitivity to estimation errors in expected returns and covariance matrices, especially in high dimensions (e.g. [4, 45]). Moreover, variance-based risk measures treat upside and downside fluctuations symmetrically, which can obscure

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tail risk exposure. In response, downside-focused risk metrics such as Value at Risk (VaR) and Conditional Value at Risk (CVaR) have gained prominence. VaR defines a loss threshold at a specified confidence level (e.g. [2, 3]), while CVaR measures the expected loss beyond that threshold (e.g. [10, 33, 62, 63, 64]). Expectile-based VaR (EVaR), introduced by Kuan et al. [46], accounts for both the magnitude and frequency of extreme losses, with nonparametric estimation (e.g. [48, 74]), and applications in financial (e.g. [9, 78]). Despite these advances in risk measurement, efficiently integrating realistic portfolio constraints—such as sparse holding rules—into high-dimensional optimization remains difficult. To address this gap, we propose a practical framework that embeds weight constraints and the SCAD penalty into CVaR-based portfolio optimization, aiming to produce sparse and robust allocations.

Modeling asset returns presents further challenges. Return series exhibit stylized facts including time-varying volatility, heavy tails, and asymmetry, which complicate risk assessment. Extreme Value Theory (EVT) provides a robust framework for capturing extreme tail risks using tools like the Generalized Pareto Distribution (GPD) (e.g. [49, 54, 55]). Because EVT assumes independently and identically distributed (i.i.d.) observations, researchers commonly filter volatility with GARCH-type models before applying EVT (e.g. [11, 25]). Consequently, hybrid GARCH-EVT frameworks have proven effective for modeling extreme market events (e.g. [12, 15, 65]). Building on this approach, we integrate EVT with an ARMA-GARCH specification to characterize the marginal distributions of asset returns, thereby explicitly capturing their tail risk properties within our optimization framework.

In high-dimensional portfolios, joint tail dependence can greatly amplify losses during crises (e.g. [16]). To model such nonlinear and asymmetric tail dependence while keeping marginal tail models separate, we use copula methods. Pair-copula constructions and regular vines (R-vines) decompose a multivariate copula into tractable bivariate building blocks, making high-dimensional dependence modeling feasible (e.g. [1, 7, 41]). Recent advances address selection and sparsity in vine structures, improving their scalability for large systems (e.g. [14, 22, 59]). Accordingly, we incorporate R-vine copulas into our integrated ARMA–GARCH-EVT framework to capture the complex dependence structures inherent in high-dimensional portfolios.

A growing body of empirical work applies hybrid GARCH-EVT-copula frameworks to capture tail events and cross-market linkages in different markets and asset classes (e.g. [36, 39, 72]). These studies validate the core pipeline for stress tests (e.g. [21, 49]), risk spillover (e.g. [44]), and scenario simulation (e.g. [65]). While these studies validate the core framework's effectiveness, they typically do not concurrently address the high-dimensional portfolios with sparsity and robustness. Our work bridges this gap by embedding a comprehensive ARMA-GARCH-EVT-R-vine copula model within a constrained and penalized high-dimensional CVaR optimization, aiming to achieve superior sparsity, stability, and out-of-sample performance.

To this end, our study implements the following integrated procedure. First, we employ an ARMA-GARCH model with t-distributed innovations to capture return dynamics and time-varying volatility. Second, we apply EVT to refine the marginal distributions of the standardized residuals, explicitly modeling the tails. Third, we utilize R-vine copulas to capture the complex, high-dimensional dependence structure among asset returns. These components together form an integrated ARMA-GARCH-EVT-R-vine copula model, which we use to generate one-day-ahead return forecasts. Finally, these forecasts feed into a high-

dimensional CVaR portfolio optimization model that incorporates weight constraints and a SCAD penalty to promote sparsity and robustness, completing the end-to-end framework.

Portfolio optimization was long confined to low-dimensional settings because of computational limits. To improve tractability, Fan et al. [28] introduce gross-exposure constraints building on [27], with numerical solutions via LARS–LASSO approach (e.g. [24]). More recent work supports high-dimensional diversification (e.g. [4, 18, 31]), but reliable estimation of expected returns and covariance matrices remains a key concern (e.g. [17, 19, 38, 52]). High-dimensional portfolios also raise trading and monitoring costs (e.g. [34]), motivating sparse holdings (e.g. [13, 23, 30, 58]). Penalization is a common remedy. Common approaches include L_1 and L_2 norms (e.g. [20, 47]), elastic nets (e.g. [73, 76]), and the smoothly clipped absolute deviation (SCAD) penalty (e.g. [26]). SCAD is particularly advantageous as it satisfies the criteria of unbiasedness, sparsity, and continuity [26], and has shown empirical promise in risk-managed portfolios (e.g. [51, 71]). However, the SCAD penalty function, expressed as the difference between two convex functions, is inherently non-convex. We address this non-convexity by adopting a penalized quantile-regression approach, which reformulates the optimization into a tractable form (e.g. [69]).

The main contributions of this study are fourfold. First, we improve large-portfolio forecasting, tail modeling, and portfolio sparsity and robustness by combining ARMA–GARCH, EVT, and R-vine copulas with SCAD regularization. Second, we resolve non-convex optimization challenges in SCAD-penalized CVaR problems by reformulating them as penalized quantile regressions solved via adaptive-LASSO algorithms. Third, we decompose the empirical gains and quantify the contribution of each model component to portfolio performance. Fourth, we benchmark our strategy against standard alternatives—equal-weighted (EWP), mean-variance (MVO), and minimum-variance (MVP)—on five dimensions: sparsity, risk exposure (VaR and CVaR), cumulative returns, risk-adjusted performance (RAROC and Sharpe), and computational efficiency.

Our sample covers the period from January 4, 2010 to December 31, 2020, with the first seven years used for training and the last four years for testing. We examine two asset universes: ten Shanghai Stock Exchange (SSE) sector indices and 131 CSI 300 constituents (filtered for data integrity). To validate the R-vine copula's ability to capture multivariate dependence, we construct and visualize a 9-layer tree structure for the ten SSE sectors. This clearly reveals sector-level connectedness and potential risk transmission, confirming the model's flexibility in representing nonlinear, asymmetric dependence in high-dimensional settings. Using the 131 stocks, we then build an integrated empirical framework that links stock-price prediction to portfolio optimization.

Empirical results show that our approach significantly increases portfolio sparsity, reduces risk (VaR and CVaR at 90%, 95%, and 99% quantile levels), and improves cumulative returns over 60-day, 90-day, and full holding periods. Risk-adjusted performance, measured by RAROC and the Sharpe ratio, is consistently superior. We compare the RAROC time series at the 90%, 95%, and 99% quantiles levels for 60- and 90-day holding periods. Across all confidence levels and both horizons, the RAROC path of our strategy consistently exceeds those of the benchmark models. Descriptive statistics substantiate this finding, indicating the strategy's superior risk-adjusted performance and robustness across market cycles. We also conduct multiple comparisons using the Scheffé test. Results show that, relative to all benchmark models, our strategy delivers statistically significant improvements in three

key metrics—RAROC, Sharpe ratio, and cumulative return—supporting the strategy's consistent outperformance. Finally, we organize the empirical workflow into three stages and report the computation time for each. The measured runtimes demonstrate that the proposed strategy is computationally feasible within realistic resource limits, confirming both its empirical effectiveness and practical viability. Collectively, this multidimensional validation establishes the advantages of our approach in terms of sparsity, risk-return efficiency, robustness, and computational efficiency.

The remainder of the paper is organized as follows. Section 2 details the methodology, including the ARMA-GARCH model, extreme value theory, and copula models. Section 3 details the practical implementation of regularization in CVaR optimization. Section 4 outlines the complete iterative procedures utilized. Section 5 describes the dataset. Section 6 presents the empirical findings, and Section 7 concludes.

2. Forecast modeling framework

Our forecasting framework has two stages. First, we model each asset's marginal return distribution using an ARMA-GARCH specification for the conditional mean and variance, with EVT applied to the tails. Second, we model the joint dependence across all assets with an R-vine copula, which captures accommodates complex, nonlinear linkages. The resulting ARMA-GARCH-EVT-R-vine model produces one-day-ahead joint return forecasts for the high-dimensional portfolio optimization.

2.1. The marginal model

The ARMA-GARCH model provides a robust framework for capturing the dynamic mean and time-varying volatility of financial returns. Following established studies [49, 65, 44, 72], we specify an ARMA(1,1) process for the conditional mean $\mu_{j,t}$ and a GARCH(1,1) process for the conditional variance $\sigma^2_{j,t}$, forming the ARMA(1,1)-GARCH(1,1):

$$\begin{aligned} r_{j,t} &= \mu_{j,t} + \phi_j(r_{j,t-1} - \mu_{j,t}) + \theta_j \varepsilon_{j,t-1} + \varepsilon_{j,t}, \\ \sigma^2_{j,t} &= \omega_j + \alpha_j \varepsilon^2_{j,t-1} + \beta_j \sigma_{j,t-1}, \\ \varepsilon_{j,t} &= z_{j,t}\sqrt{\sigma_{j,t}}, \quad z_{j,t} \approx (i.i.d.), \end{aligned} \tag{1}$$

where $r_{j,t}$ denotes the actual returns for the j-th asset ($j = 1, 2, \ldots, p$), and $z_{j,t}$ represents the standardized residuals. The parameters ϕ_j and θ_j represent the autoregressive (AR) and moving average (MA) dynamics of the conditional mean, respectively, satisfying $\phi_j + \theta_j \neq 0$. Furthermore, α_j captures the asymmetry of past returns on volatility, β_j accounts for the GARCH dynamics of the conditional variance, and ω_j represents the weighted long-run variance. These parameters are subject to the constraints $\omega_j > 0$, $\alpha_j \geq 0$, $\beta_j \geq 0$, and $\alpha_j + \beta_j < 1$.

To model extreme tails, we apply EVT to the standardized residuals $\{\hat{z}_{j,t}\}$ from the fitted ARMA–GARCH models. Using the peaks-over-threshold (POT) method, we fit Generalized Pareto Distributions (GPDs) to exceedances above an upper threshold u^U_j and below a lower threshold u^L_j. Values in the central region are modeled with a Gaussian kernel density

estimator. The resulting semi-parametric marginal CDF for asset j is

$$F_j(\hat{z}_j) = \begin{cases} \dfrac{T_{u_j^L}}{N}\left(1+\xi_j^L\dfrac{u_j^L-\hat{z}_j}{\beta_j^L}\right)^{-1/\xi_j^L}, & \text{for } \hat{z}_j < u_j^L; \\ \Phi_j(\hat{z}_j), & \text{for } u_j^L \le \hat{z}_j \le u_j^U; \\ 1-\dfrac{T_{u_j^U}}{N}\left(1+\xi_j^U\dfrac{\hat{z}_j-u_j^U}{\beta_j^U}\right)^{-1/\xi_j^U}, & \text{for } \hat{z}_j > u_j^U, \end{cases} \tag{2}$$

where $T_{u_j^L}$ and $T_{u_j^U}$ count exceedances below and above the thresholds, N is the sample size, and (ξ_j, β_j) are the GPD shape and scale parameters. This hybrid EVT-kernel approach yields smooth marginals with accurate tail description. With continuous marginal CDFs $F_j(\cdot)$ $(j = 1, \ldots, p)$ in hand, we transform residuals to uniform scores $u_{j,t} = F_j(\hat{z}_{j,t})$ for dependence modeling.

2.2. The dependence structure model

Let random returns vector $\mathbf{r} = (\mathbf{r}_1, \ldots, \mathbf{r}_p)$ has joint CDF F and corresponding marginal CDFs F_j $(j = 1, \ldots, p)$. By Sklar's theorem [66] and uniform transformations $U_j = F_j(\mathbf{r}_j)$, there exists a copula C such that

$$F(r_1, \ldots, r_p) = C(u_1, \ldots, u_p) = \mathbb{P}(U_1 \le u_1, U_2 \le u_2, \ldots, U_p \le u_p), \tag{3}$$

where $u_j = F_j(r_j)$, and $\mathbb{P}(\cdot)$ denotes the probability of the corresponding event. The copula C fully captures the dependence structure among the variables, each margin following a uniform distribution.

If F_j $(j = 1, \ldots, p)$ and C are continuous derivatives, the joint probability density function (PDF) f can be expressed via the copula density c [66]:

$$f(r_1, \ldots, r_p) = c(u_1, \ldots, u_p)\prod_{i=1}^{p} f_i(r_i), \quad \text{where} \quad c(u_1, \ldots, u_p) = \frac{\partial^p C(u_1, \ldots, u_p)}{\partial u_1 \cdots \partial u_p}, \tag{4}$$

where f_i is the marginal PDF for random variable $\mathbf{r}_i$.

Conditional copulas provide a further decomposition of each constituent term within this product. Take, for instance, the bivariate density $f(r_1, r_2)$; it can be detailed as the product of a copula density and marginal densities: $c_{1,2}(F_1(r_1), F_2(r_2))f_1(r_1)f_2(r_2)$. This expands our understanding to define $f(r_1|r_2) = c_{1,2}(F_1(r_1), F_2(r_2))f_1(r_1)$. In a similar manner, we can further delineate the subsequent factors in the equation, with the exception of the initial term, using the following relationship:

$$f(r|v) = c_{r,v_j|\mathbf{v}_{-j}}\left[F\left(r|\mathbf{v}_{-j}\right), F\left(v_j|\mathbf{v}_{-j}\right)\right] \cdot f\left(r|\mathbf{v}_{-j}\right). \tag{5}$$

Here, $\mathbf{v}$ signifies an n-dimensional vector where n ranges from 1 to p, v_j is an arbitrarily selected component of $\mathbf{v}$, and $\mathbf{v}_{-j}$ refers to the vector $\mathbf{v}$ with the omission of the element v_j. Moreover, the bivariate copula density function is specified by $c_{x,v_j|\mathbf{v}_{-j}}$, and $F(\cdot|\cdot)$ symbolizes the conditional distribution function.

Furthermore, while conventional multivariate copulas suggest a homogenous dependency structure across all variables, vine copulas permit modeling of differential pairwise dependencies, enhancing the flexibility and precision in capturing inter-variable relationships within higher-dimensional contexts [1, 7, 42]. Vine copulas thereby enable the construction of a more tailored and sophisticated modeling of dependencies that better reflects the complexities inherent in financial data.

Vines, as introduced by Bedford & Cooke [8], facilitate complex decompositions of multivariate dependencies. A vine $\mathbf{V}$ for p variables consists of a series of nested trees $\mathbf{V} = \{T_1, \ldots, T_{p-1}\}$, where the edges of tree T_j transform into nodes in tree T_{j+1}. Kurowicka & Cooke [50] define a regular vine in such a way that two edges in tree T_j can be connected in tree T_{j+1} only if they have a shared node. Dissmann et. al. [22] extend regular vine copula applicability to high-dimensional finance through generalized density evaluation and graph-based automated selection, revealing critical interdependencies in equity, bond, and commodity markets.

Definition 1. *[50] (**Vine, regular vine**)* $\mathbf{V}$ *is a vine on n elements if*

(1) $\mathbf{V} = \{T_1, \ldots, T_{p-1}\}$,

(2) *T_1 is a connected tree where the nodes $N_1 = \{1, \ldots, n\}$ are joined by edges E_1. For $j = 2$ to $n-1$, tree T_j is constructed from nodes $N_j = E_{j-1}$.*

$\mathbf{V}$ *is called a regular vine if in addition it satisfies the following proximity condition:*

(3) *For any $j = 2$ to $n-1$, given an edge $\{a, b\} \in E_j$, it holds that $|(a \triangle b)| = 2$, utilizing $\triangle$ to indicate symmetric difference between sets and $|\cdot|$ for set cardinality.*

In a tree T_j within a vine structure, the order of an edge is labeled as $j-1$, where j ranges from 1 to $n-1$. The degree of a node is the total count of its connected edges. A regular vine possesses a unique form called a C-vine or Canonical Vine if, in each tree T_j, there exists exactly one node with a degree of $n-j$. On the other hand, if every node in the initial tree T_1 has a degree of 2 at most, the vine qualifies as a D-vine or Drawable Vine.

Definition 2. *[50] (Canonical Vine, Drawable Vine) A regular vine is called a*

(i) *C-vine if each tree T_j has a unique node of degree $N-j$, and hence has the maximum degree.*

(ii) *D-vine if all nodes in T_1 have degrees no higher than 2.*

Bedford & Cooke [7, 8] and Kurowicka & Cooke [50] suggest that an edge within an R-vine, denoted as $e \in E_i$, can be determined by a conditioned node pair $(j(e), k(e))$ combined with a conditioning set $D(e)$, represented by $e = (j(e), k(e)|D(e))$. Consequently, this informs the definition of the corresponding pair-copula as $c_{j(e),k(e)|D(e)}$. Assuming we are provided with marginal distributions F_i and their densities $f_i, i = 1, \ldots, n$, the density of an R-vine copula may then be depicted as a product combining conditional and marginal copula associated with these edges:

$$f_{1\ldots,n} = \prod_{k=1}^{p} f_k \prod_{i=1}^{p-1} \prod_{e \in E_i} c_{j(e),k(e)|D(e)} \left(F_{j(e)|D(e)}, F_{k(e)|D(e)} \right). \tag{6}$$

In this study, we employ R-vine copulas to model high-dimensional dependence using the `RVineStructureSelect` function in R. The structure selection follows the maximum spanning tree algorithm with likelihood-based edge selection, as proposed by [14, 22]. The fitted R-vine, together with the marginal ARMA–GARCH–EVT models, is used to simulate joint residuals and produce one-step-ahead return forecasts. These resulting forecasts are then used as inputs for the high-dimensional portfolio optimization model, which incorporates a SCAD penalty to promote sparsity and robustness, as detailed in the following section.

3. Construction of high-dimensional portfolio optimization

This section presents a portfolio optimization model for high-dimensional settings using CVaR as the risk measure. We first outline the theoretical foundation of CVaR and then introduce a regularization framework that incorporates the SCAD penalty proposed by Fan & Li [26]. This combination promotes sparsity, improves robustness, and strengthens out-of-sample performance.

3.1. CVaR *risk measure*

Let Y denote portfolio return and $\tau \in (0,1)$ be the significance level (the tail probability). CVaR at level τ is defined as the conditional expectation of losses:

$$\mathrm{CVaR}_\tau(Y) = -\mathbb{E}\big[Y \mid Y \le -\mathrm{VaR}_\tau\big], \tag{7}$$

where VaR_τ is the τ-quantile of the loss distribution. By applying the definition of conditional expectation, CVaR can be expressed in integral form as:

$$\mathrm{CVaR}_\tau(Y) = -\int_{-\infty}^{+\infty} yp\left(y|y \le -\mathrm{VaR}_\tau\right) dy, \tag{8}$$

where $p(\cdot|\cdot)$ denotes the conditional probability density. Using Bayes' theorem, we have

$$p(y \mid Y \le -\mathrm{VaR}_\tau) = \frac{p(y)\,\mathbb{I}(y \le -\mathrm{VaR}_\tau)}{\mathbb{P}(Y \le -\mathrm{VaR}_\tau)}, \tag{9}$$

where $p(y)$ is the marginal density of Y, $\mathbb{I}(\cdot)$ is the indicator function, and $\mathbb{P}(Y \le -\mathrm{VaR}_\tau) = \tau$. Substituting this into the integral representation yields:

$$\mathrm{CVaR}_\tau(Y) = -\frac{1}{\tau}\int_{-\infty}^{+\infty} y\,p(y)\,\mathbb{I}(y \le -\mathrm{VaR}_\tau)\,dy, \tag{10}$$

By observing that the inequality $y \le -\mathrm{VaR}_\tau$ is equivalent to the cumulative probability condition $0 \le F_Y(y) \le \tau$, we perform the change of variable $t = F_Y(y)$. This yields:

$$\mathrm{CVaR}_\tau(Y) = -\frac{1}{\tau}\int_0^\tau F_Y^{-1}(t)F_Y'(y)dF_Y^{-1}(t) = -\frac{1}{\tau}\int_0^\tau F_Y^{-1}(t)dt, \tag{11}$$

where $F_Y(\cdot)$ is the CDF of Y, and $F_Y^{-1}(\cdot)$ is the inverse CDF (quantile function). The second equation follows from the relationship $dF_Y^{-1}(t) = \frac{1}{F_Y'(y)}dt$. The expression (11) clearly shows

that CVaR measures the average of the worst τ-quantile of losses, offering a more complete and informative view of tail risk than VaR alone. This formulation is especially useful in portfolio optimization and risk management, as it accounts not only for the threshold of extreme losses but also for their expected magnitude beyond that threshold.

3.2. High-dimensional portfolio with SCAD penalty

Consider a portfolio of p assets. Let $\boldsymbol{r} = (r_1, \ldots, r_p)^\top$ denote the vector of asset returns and let $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^\top$ be the vector of portfolio weights. The portfolio return is $Y = \boldsymbol{r}^\top \boldsymbol{\beta}$, and the corresponding loss is $-Y$. To obtain sparse and robust allocations in a high-dimensional setting, we incorporate a SCAD penalty into the CVaR optimization. The resulting model is specified as follows:

$$\begin{aligned} \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \ & \mathrm{CVaR}_\tau(-\boldsymbol{r}^T \boldsymbol{\beta}) \\ \text{s.t. } & \mathrm{Pen}_\lambda(\boldsymbol{\beta}) \leq \bar{\omega}, \\ & \mathbf{e}^T \boldsymbol{\beta} = 1, \end{aligned} \tag{12}$$

where $\mathbf{e}$ is a p-vector of ones, $\mathrm{Pen}_\lambda(\cdot)$ is the SCAD penalty, $\lambda > 0$ is the penalty level, and $\bar{\omega}$ bounds the total penalty.

The i-th component of SCAD penalty, denoted as $p_\lambda^{\mathrm{scad}}(\beta_i)$ for β_i, is defined as [26]:

$$p_\lambda^{scad}(\beta_i) = \begin{cases} \lambda|\beta_i|, & \text{for } |\beta_i| \leq \lambda \\ -\frac{|\beta_i|^2 - 2\alpha\lambda|\beta_i| + \lambda^2}{2(\alpha-1)}, & \text{for } \lambda < |\beta_i| \leq \alpha\lambda \\ \frac{(\alpha+1)\lambda^2}{2}, & \text{for } |\beta_i| > \alpha\lambda, \end{cases} \tag{13}$$

with tuning parameter $\alpha > 2$. This formulation can be construed as a quadratic spline with knots located at λ and $\lambda\alpha$. For the full weight vector $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)$, the total SCAD penalty is the sum over all components:

$$\begin{aligned} \mathrm{Pen}_\lambda(\boldsymbol{\beta}) := \sum_{i=1}^p p_\lambda^{scad}(\beta_i) = & \sum_{i=1}^p \lambda|\beta_i| \times \mathbb{I}(|\beta_i| \leq \lambda) + \sum_{i=1}^p \tfrac{(\alpha+1)\lambda^2}{2} \times \mathbb{I}(|\beta_i| > \alpha\lambda) \\ & - \sum_{i=1}^p \left(\tfrac{|\beta_i|^2 - 2\alpha\lambda|\beta_i| + \lambda^2}{2(\alpha-1)} \right) \times \mathbb{I}(\lambda < |\beta_i| \leq \alpha\lambda), \end{aligned} \tag{14}$$

where $\mathbb{I}(\cdot)$ is the indicator function.

The SCAD penalty induces three types of shrinkage.

(1) Small coefficients ($|\beta_i| \leq \lambda$) are shrunk to zero, achieving variable selection.

(2) Moderate coefficients ($\lambda < |\beta_i| \leq \alpha\lambda$) are partially shrunk toward zero.

(3) Large coefficients ($|\beta_i| > \alpha\lambda$) remain essentially unpenalized, preserving unbiasedness.

This three-fold facilitates the creation of a sparse solution that not only reduces complexity but also aims to maintain unbiased estimates for meaningful coefficients.

The constrained optimization formulation (12) can be equivalently expressed as the regularized problem through Lagrangian duality, where the penalty parameter λ implicitly

enforces the constraint $\text{Pen}_\lambda(\boldsymbol{\beta}) \leq \bar{\omega}$, thereby enhancing computational tractability:

$$\begin{aligned} \min_{\boldsymbol{\beta}\in\mathbb{R}^p} \quad & \text{CVaR}_\tau(-\boldsymbol{r}^T\boldsymbol{\beta}) + \text{Pen}_\lambda(\boldsymbol{\beta}) \\ s.t. \quad & \mathbf{e}^T\boldsymbol{\beta} = 1. \end{aligned} \tag{15}$$

4. Algorithm for high-dimensional portfolio optimization

Building on the framework of Bassett et al. [5], the regularized high-dimensional portfolio optimization problem (15) can be reformulated as a penalized quantile regression problem. This equivalence relies on the quantile check function:

$$\rho_\tau(u) = u\left(\tau - \mathbb{I}(u < 0)\right). \tag{16}$$

4.1. The penalized quantile regression problem

The connection between CVaR minimization and quantile regression is established by the following theorem.

Theorem 1. *For a random return Y with finite expectation $\mu = \mathbb{E}[Y]$, the τ-level CVaR can be expressed as*

$$\text{CVaR}_\tau(Y) = \frac{1}{\tau}\,\mathbb{E}\left[\rho_\tau(Y - \xi_\tau)\right] - \mu, \tag{17}$$

where $\xi_\tau = F_Y^{-1}(\tau)$ is the τ-quantile of the distribution of Y.

Proof. The proof is provided in Appendix A. □

Theorem 1 clarifies that minimizing CVaR is equivalent to minimizing the expected check-loss of the return relative to its τ-quantile, scaled by $1/\tau$ and adjusted for the mean return.

Given that $\boldsymbol{\beta}$ satisfies the budget constraint $e^T\boldsymbol{\beta} = 1$, we define the portfolio return Y as follows

$$Y = \mathbf{r}^T\boldsymbol{\beta} = r_1 + \textstyle\sum_{i=2}^{p}(r_i - r_1)\beta_i.$$

Substituting this expression into Equation (17) transforms the regularized CVaR optimization problem (15) into the following penalized quantile regression formulation:

$$\min_{\beta\in\mathbb{R}^p} \frac{1}{\tau}\mathbb{E}\left[\rho_\tau(r_1 + \sum_{i=2}^{p}(r_i - r_1)\beta_i - \xi_\tau)\right] + \text{Pen}_\lambda(\boldsymbol{\beta}), \tag{18}$$

where $\text{Pen}_\lambda(\boldsymbol{\beta})$ is the SCAD penalty defined in (14).

To solve problem (18), we apply the adaptive-LASSO penalized quantile regression method proposed by Wu & Liu [69]. It merges quantile regression—modeling conditional return quantiles—with the adaptive-LASSO penalty, which enforces sparsity in portfolio weights. Our integrative approach facilitates the selection and estimation of important assets for portfolio optimization, simultaneously managing the preferred risk exposure as quantified by the CVaR.

4.2. Forecasting methodology

We implement a rolling-window scheme of fixed length L. Starting at time t_0, a one-step-ahead forecast for t_0+1 is produced using data from the window $[t_0-L+1, t_0]$. The window then shifts forward one period, and the process is repeated throughout the forecast horizon T. The complete forecasting and optimization procedure consists of the following seven steps:

Step 1. Fit ARMA-GARCH models: For each asset $i = 1, \ldots, p$, estimate the ARMA(1,1)-GARCH(1,1) model (1) on the window $t = t_0 - L + 1, \ldots, t_0$. Obtain the standardized residuals

$$\hat{z}_t = (\hat{z}_{1t}, \hat{z}_{2t}, \ldots, \hat{z}_{pt}), \qquad t = t_0 - L + 1, \ldots, t_0, \tag{19}$$

which are treated as approximately i.i.d. for each asset.

Step 2. Estimate marginal distributions via EVT: Apply the POT method (2) to each residual series $\{\hat{z}_{it}\}$. Fit a GPD to the tail exceedances and a Gaussian kernel density to the center. This yields the estimated marginal distribution $\hat{F}_i(\cdot)$ for each asset. Transform the residuals to uniform variates:

$$\hat{u}_{it} = \hat{F}_i(\hat{z}_{it}) \sim U(0,1), \qquad i = 1, \ldots, p;\ t = t_0 - L + 1, \ldots, t_0. \tag{20}$$

Step 3. Fit an R-vine copula: Using the uniform vectors $\hat{\boldsymbol{u}}_t = (\hat{u}_{1t}, \ldots, \hat{u}_{pt})$, select an R-vine structure, choose a bivariate copula family for each pair, and estimate the corresponding parameters. The resulting copula $\hat{C}$ approximates the joint distribution:

$$\hat{F}(\hat{z}_{1t}, \ldots, \hat{z}_{pt}) = \hat{C}(\hat{u}_{1t}, \ldots, \hat{u}_{pt}), \qquad t = t_0 - L + 1, \ldots, t_0. \tag{21}$$

Step 4. Simulate from the copula: Generate K independent draws $\hat{\boldsymbol{v}}_k = (\hat{v}_{k1}, \ldots, \hat{v}_{kp})$, $k = 1, \ldots, K$, from the fitted R-vine copula $\hat{C}$. Each $\hat{v}_{kj}$ follows a uniform distribution on $[0,1]$ while preserving the estimated cross-asset dependence.

Step 5. Transform to the residual space: Apply the inverse of the estimated marginal CDFs to convert the uniform draws $\hat{\boldsymbol{v}}_k$ back to the scale of the standardized residuals:

$$\hat{\mathbf{w}}_k = (\hat{w}_{k1}, \hat{w}_{k2}, \ldots, \hat{w}_{kp}) = \left(\hat{F}_1^{-1}(\hat{v}_{k1}), \hat{F}_2^{-1}(\hat{v}_{k2}), \hat{F}_p^{-1}(\hat{v}_{k1})\right) \quad k = 1, \ldots, K. \tag{22}$$

Step 6. Generate return forecasts: The forecast for the next step, $t = t_0 + 1$, is generated by inputting the estimated residuals $\hat{\mathbf{w}}_k$ back into the ARMA-GARCH model:

$$\hat{\mathbf{r}}_t^k = \left(\hat{r}_{t1}^k, \ldots, \hat{r}_{tp}^k\right), \quad k = 1, \ldots, K. \tag{23}$$

Each $\hat{r}_{ti}^k$ corresponds to the forecast for series i.

Step 7. Solve the penalized portfolio optimization: Upon incorporating the forecasted returns, $\hat{r}_t^k$, into the optimization (18), we determine the optimal portfolio weights $\boldsymbol{\beta}_t$ by using an adaptive-LASSO penalized quantile regression to minimize expected losses

and penalty function as (18). Once $\boldsymbol{\beta}_t$ is obtained, we proceed to compute the single-period portfolio return as $R_t = \sum_{i=1}^{p} \beta_{ti}\hat{r}_{ti}$ and evaluate the CVaR employing the formula provided in Eq. (17).

Use the forecasted returns $\{\hat{\mathbf{r}}_k\}$ as inputs to the penalized quantile regression problem (18). Solve for the optimal weight vector $\boldsymbol{\beta}_t$ via adaptive LASSO-penalized quantile regression. The resulting portfolio return at t is

$$R_t = \boldsymbol{\beta}_t^{\top}\hat{\boldsymbol{r}}, \tag{24}$$

where $\hat{\mathbf{r}}$ is the mean forecast across the K simulations. The associated CVaR is computed from (17). Update the time index $t_0 = t_0 + 1$ and return to Step 1.

5. Data

This study employs daily data on Chinese assets from two distinct categories: ten sector indices from the SSE and individual equities from the CSI 300 Index. The full sample spans from January 4, 2010, to December 31, 2020, covering all trading days within this period. The dataset is divided into a 7-year in-sample period for model training (from January 2010 to December 2016) and a subsequent 4-year out-of-sample period for performance validation (from January 2017 to December 2020). To ensure robustness and simulate realistic investment conditions, we implement a rolling-window estimation and forecasting framework.

For sector-level analysis, we employ ten major SSE indices: Energy, Materials, Industrials, Consumer Discretionary, Consumer Staples, Health Care, Financials, Information Technology, Communication Services, and Utilities. Daily data for all ten sectors cover the full January 2010 – December 2020 period.

For the high-dimensional portfolio analysis, we use individual stocks from the CSI 300 Index. To ensure data quality and model stability, we conduct validation and outlier screening, removing stocks with abnormal returns, missing data, or that fail to converge during the model-fitting stage. After this filtering, 131 stocks are retained for the empirical analysis. AS a result, a daily rolling-window backtesting approach is used to simulate real-time information updating and trading decisions. All daily closing prices are sourced from the Wind Database[1]. For each asset on trading day t, the log return is computed as $r_t = \ln \frac{p_t}{p_{t-1}}$, where p_t denotes the closing price on day t.

6. Empirical results

All empirical analyses were run on a MacBook Pro equipped with a 1.4 GHz quad-core Intel Core i5 processor, 8 GB of 2133 MHz LPDDR3 RAM, and Intel Iris Plus Graphics 645.

6.1. EVT and Copula dependency structure analysis

Table 1 reports the parameter estimates for the ARMA(1,1)-GARCH(1,1) models. The estimates for the GARCH(1,1) model are provided in Appendix B. Across all specifications,

[1] https://www.wind.com.cn/portal/zh/EDB/index.html

the estimated parameters satisfy the expected theoretical constraints: the ARMA parameters (ϕ, θ) lie within the stationary region (absolute value < 1), and the GARCH coefficients (α, β) are positive. Furthermore, $\alpha + \beta$ are close to one across all sectors, confirming the high persistence of volatility. A likelihood ratio test (LRT) comparing the ARMA(1,1)-GARCH(1,1) and GARCH(1,1) shows that the ARMA(1,1)-GARCH(1,1) specification provides a significantly better fit for seven out of ten sectors (the last column of Table 1). Consequently, we adopt the ARMA(1,1)-GARCH(1,1) model for subsequent analysis.

Table 1: ARMA(1,1)-GARCH(1,1) parameters for daily log returns of sector indexes.

Series	μ	ϕ	θ	ω	α	β	λ	LRT
Energy	-1.66e-04	-0.885	0.902	3.12e-06	0.056	0.933	1.177	3.984(0.136)
Materials	3.14e-04	0.190	-0.165	3.96e-06	0.066	0.921	1.332	1.801(0.406)
Industrials	2.86e-05	-0.826	0.852	2.98e-06	0.063	0.924	1.247	6.003(0.050)**
Cons Disc	4.41e-04	-0.826	0.855	1.16e-06	0.065	0.933	1.369	7.473(0.024)**
Cons Staples	7.74e-04	-0.849	0.873	3.80e-06	0.072	0.913	1.482	5.610(0.061)*
Health Care	6.23e-04	-0.725	0.775	8.57e-07	0.061	0.938	1.524	14.190(0.001)***
Financial	-1.79e-04	0.831	-0.861	2.12e-06	0.063	0.930	1.071	10.009(0.007)***
Info Technology	3.82e-04	-0.821	0.860	2.32e-06	0.049	0.946	1.436	12.377(0.002)***
Communication Services	5.30e-04	-0.839	0.872	5.63e-06	0.062	0.922	1.370	9.533(0.009)***
Utilities	4.99e-06	-0.860	0.873	1.03e-06	0.066	0.930	1.209	1.884(0.390)

Note: The total number of observations used for each stock index is 2674.

We then apply EVT to the standardized residuals from the ARMA-GARCH models. Using the POT method, we fit the GPD to the tails. The estimated thresholds (μ), shape (ξ), and scale (β) parameters are reported in Appendix B. To model the high-dimensional dependence across the ten sector indices, we adopt an R-vine copula. The sectors are coded numerically as follows: 1 Energy, 2 Materials, 3 Industrials, 4 Consumer Discretionary, 5 Consumer Staples, 6 Health Care, 7 Financials, 8 Information Technology, 9 Communication Services, 10 Utilities. The R-vine structure is selected automatically using the `RVineStructureSelect` function in R, following the algorithm of Brechmann et al. [14] and Dissmann et. al. [22]. The procedure operates in three stages:

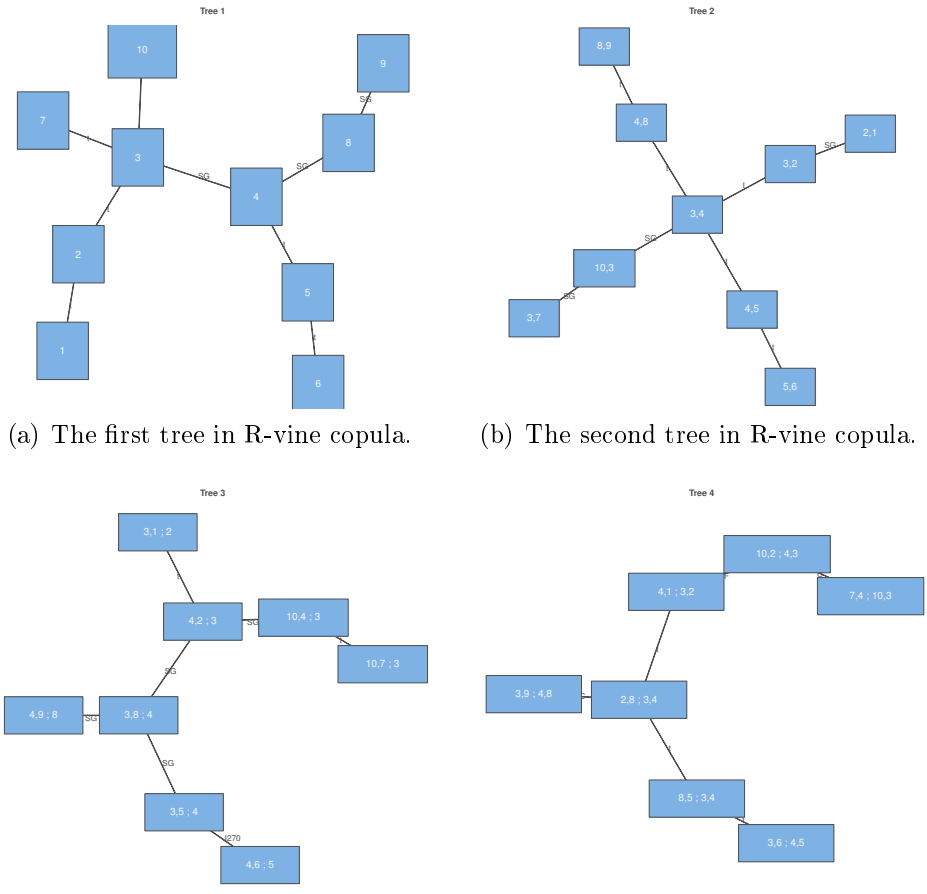
- Tree selection and ordering: the vine tree structure and edge sequence are determined by minimizing the Akaike Information Criterion (AIC).
- Copula family selection: for each edge, the optimal bivariate copula family is chosen from a pre-specified set via pairwise AIC comparison. selects optimal copula families from predefined candidates via pairwise AIC evaluation.
- Independence pruning: edges whose dependence is statistically insignificant are removed, simplifying the overall dependence graph.

In this study, the R-vine tree selection employs maximum spanning trees with edge weights set as the absolute empirical Kendall's tau, say $\hat{\tau}_{ij}$. For each tree, it solves:

$$\max \sum_{edges(i,j) \in T_\ell} \hat{\tau}_{ij}, \tag{25}$$

where T_ℓ forms a spanning tree on the node set. The first tree is always constructed from a complete graph. Subsequent trees adhere to R-vine proximity conditions, which constrain allowable edges based on the existing structure.

Figure 1 graphically displays the first four trees of the selected R-vine structure, which provide a valuable contribution to our understanding of the existing interdependencies and assist in understanding how R-vine copula effectively represents the dependence structures in multivariate random variables. The first tree (Figure 1(a)) reveals the strongest pairwise dependencies, predominantly modeled by the t-copula and the survival Gumbel copula. Most dependencies are symmetric (well-described by the t-copula), while the survival Gumbel copula identifies three pairs with clear asymmetry. The t-copula is particularly suitable for financial returns because it captures both upper and lower tail dependence—a crucial feature for modeling joint extreme movements during market stress or crashes. Subsequent trees (Trees 2–9) model conditional dependencies and employ a wider variety of copula families (e.g., Clayton, Frank), demonstrating the flexibility of the vine approach in capturing a complex, high-dimensional dependence structure. Trees 5–9 are presented in Appendix B.



(a) The first tree in R-vine copula.

(b) The second tree in R-vine copula.

(c) The third tree in R-vine copula.

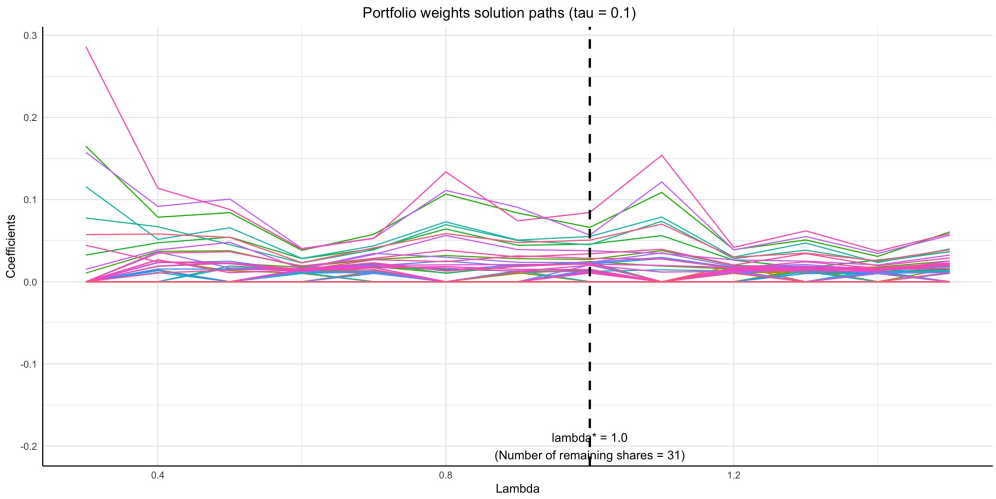
(d) The fourth tree in R-vine Copula

Figure 1: An illustative representation of R-vine copulas structures for Tree 1–Tree 4.

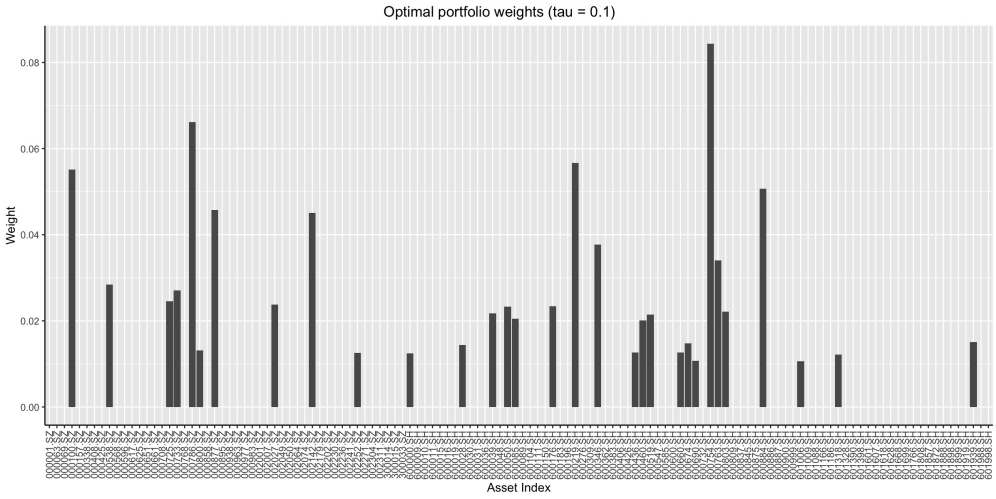
6.2. Sparsity and robustness analysis

The forecasted returns from the integrated ARMA-GARCH-EVT-R-vine model are used as inputs to the SCAD-penalized high-dimensional portfolio optimization model (12). We first examine how the SCAD penalty induces sparsity in portfolio weights. Although we discuss generalized cross-validation (GCV) for selecting SCAD parameters (λ, α) in Appendix C, empirical evidence indicates that model sparsity is largely insensitive to their specific values. We therefore fix $\lambda = 1$ and $\alpha = 3.7$ in the main analysis. Empirical results show that the SCAD-based portfolio maintains strong sparsity and robustness without excessive fine-tuning (λ, α). This also demonstrates that the model's effectiveness stems from its inherent statistical design, not from extensive parameter optimization.

Figure 2 illustrates this sparsity-inducing effect. Panel 2(a) shows the solution paths: as the penalty parameter λ increases, the weights of most assets shrink rapidly to zero. Panel 2(b) displays the optimal weights for $\lambda = 1$ and $\tau = 0.1$. Its horizontal axis represents stock codes, and the vertical axis indicates the respective investment weights of these stocks. The resulting portfolio is highly sparse, with the majority of weights being zero. No single asset weight exceeds 0.25, ensuring diversification among the selected holdings. This yields a concentrated yet diversified portfolio that is easier to manage and aligns with a risk-averse strategy. Additional tests at $\tau = 0.01$ and 0.05 (see Appendix C.1) confirm that sparsity and robustness are maintained across different risk levels CVaR_τ.



(a) Portfolio weight paths at $\tau = 0.1$.



(b) Portfolio weights at $\lambda = 1$ and $\tau = 0.1$.

Figure 2: Portfolio weight paths at $\tau = 0.1$ and weights at $\lambda = 1$.

6.3. Risk analysis across portfolios

We compare the risk performance of our SCAD-based strategy with three benchmarks: the Equal-Weighted Portfolio (EWP), the Minimum-Variance Portfolio (MVP), and the Mean-Variance Optimization (MVO). Figure 3 plot the out-of-sample CVaR at 99% confidence level over the evaluation period. Our SCAD strategy consistently produces the lowest VaR and CVaR, indicating superior tail risk management. The EWP exhibits the highest risk, while MVP and MVO show intermediate risk profiles.

To highlight the value of modeling dependence, we compare the risk forecasts from the full model (with R-vine copula) against a version that omits the copula (without R-vine copula). As shown in Figure 4, incorporating the R-vine copula leads to a systematic reduction in both VaR and CVaR at the 99% confidence level. This improvement stems from the copula's ability to capture complex, nonlinear dependence structures among assets, which in turn yields more accurate return forecasts and a more realistic assessment of portfolio tail risk. Consequently, asset-allocation decisions based on the copula-enhanced model are

more prudent and effective in mitigating extreme losses. These findings remain robust across different risk-measure specifications, holding for CVaR at the 95% and 90% confidence levels, as well as for VaR at the 99%, 95%, and 90% levels. Detailed results are provided in Appendix C.2.

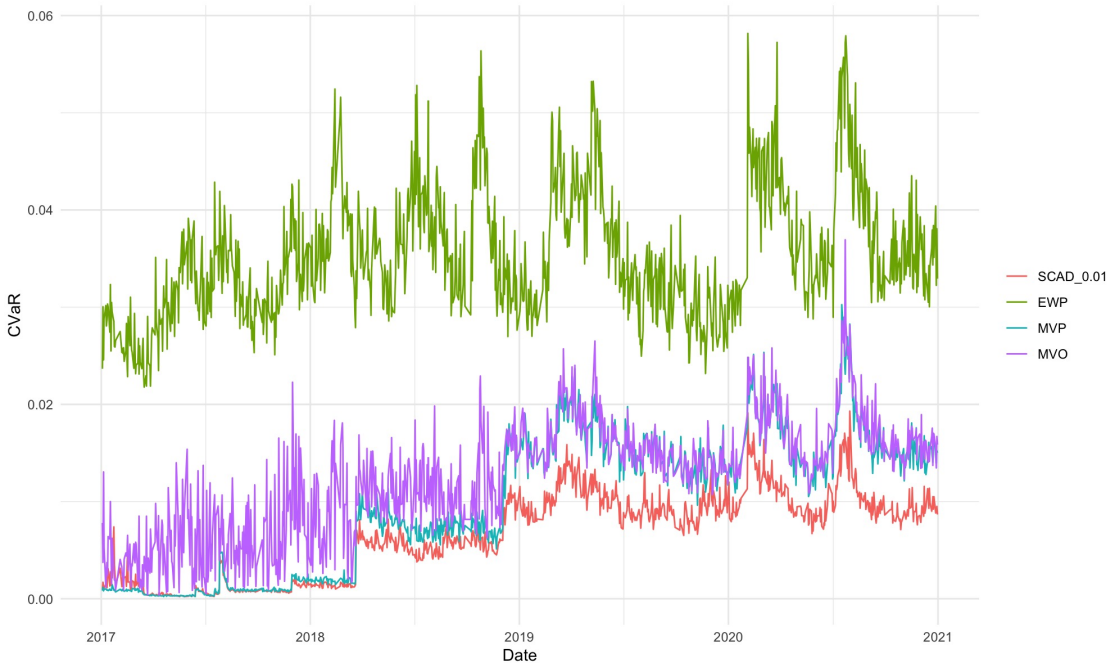


Figure 3: Comparison of CVaR at 99% confidence level across portfolios.

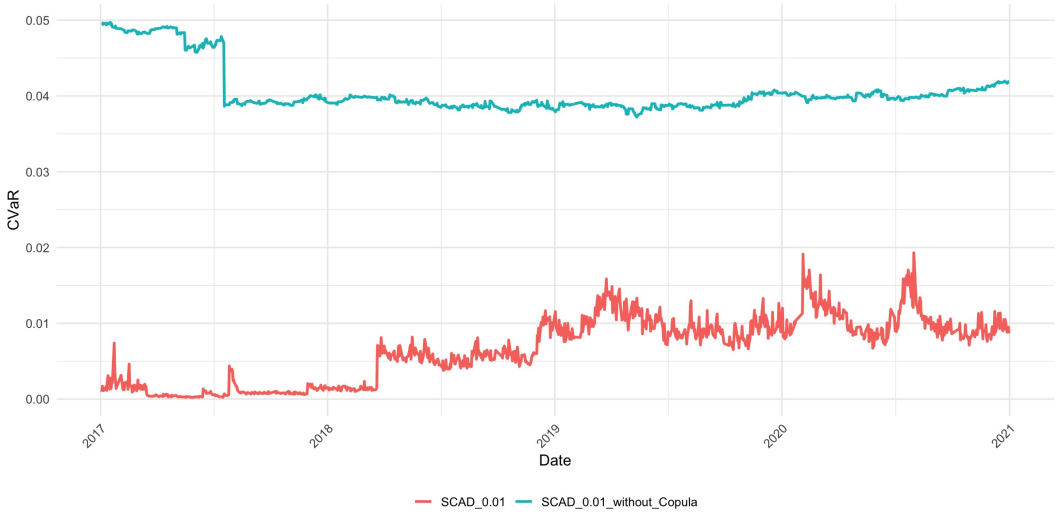


Figure 4: Forecasted CVaR at 99% confidence level: with vs. without R-vine copula.

6.4. Return analysis across portfolios

To assess how the SCAD penalty affects portfolio performance, we compare three penalty strengths: "SCAD_tau_0.01", "SCAD_tau_0.05", and "SCAD_tau_0.1", each corresponding to problem (18) with the respective $\tau = 0.01,\ 0.05$, and 0.1. We benchmark

these SCAD-based portfolios against three conventional strategies that all employ the R-vine copula for dependence modeling: EWP, MVP, and MVO.

Figure 5 compares the cumulative returns of all strategies and the CSI 300 Index during the out-of-sample period (2017–2020). All variants of our strategy provide effective return profiles: $\tau = 0.05$ delivers the highest cumulative return, followed by $\tau = 0.1$, while $\tau = 0.01$ yields the lowest. This suggests that $\tau = 0.05$ strikes an optimal balance between mitigating extreme losses and preserving return potential, whereas $\tau = 0.01$ reduces returns through excessive conservatism. Notably, during market downturns such as the 2018-2019 Sino-U.S. trade tensions, our strategy shows strong resilience—limiting drawdowns while still capturing upside rebounds once conditions stabilize. Appendix C.3 displays the cumulative return curves for portfolios constructed using a rolling-window approach, with holding periods of 60 and 90 days. The results consistently confirm the superiority of our proposed strategy.

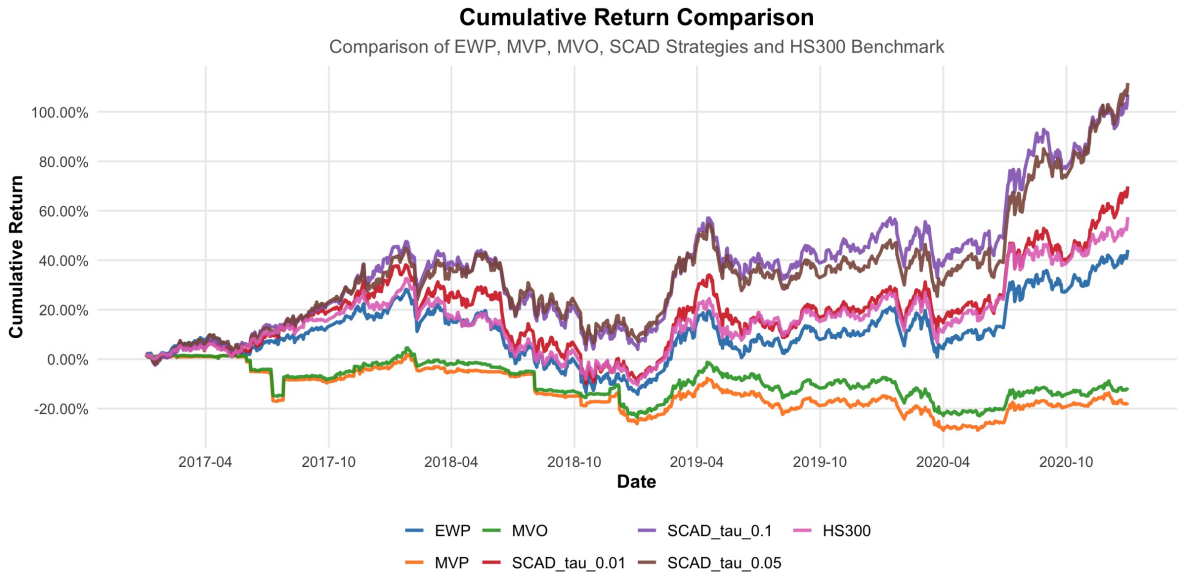


Figure 5: Comparison of cumulative return across different portfolios.

Table 2 summarizes key out-of-sample performance metrics: Cumulative returns (%), Annualized return (%), Volatility, Sharpe ratio, and Maximum drawdown (%). The SCAD strategy with $\tau = 0.05$ achieves the highest annualized return (21.41%) and Sharpe ratio (0.913), while maintaining a maximum drawdown (26.8%) below that of the market. In contrast, traditional strategies show negative Sharpe ratios (MVP Sharpe: -0.468; MVO Sharpe: -0.357).

Table 3 reports statistical tests for outperformance relative to the CSI 300 benchmark. The strategies with $\tau = 0.05$ and 0.1 show statistically significant positive excess returns ($p < 0.05$ for both the t-test and Wilcoxon signed-rank test) and economically meaningful improvements in the Sharpe ratio differential ($\triangle Sharpe$). The $\tau = 0.01$ strategy shows positive but statistically insignificant excess returns. Traditional strategies show no significant advantage.

Table 2: Out-of-sample performance metrics comparison across portfolios.

Strategies	Cumulative return	Annualized return	Volatility	Sharpe	Maximum drawdown
CSI 300	57.44	12.46	0.191	0.548	32.5
EWP	44.21	9.94	0.193	0.410	33.1
MVP	-17.50	-4.86	0.147	-0.468	30.1
MVO	-11.51	-3.11	0.143	-0.357	26.8
SCAD_tau_0.01	69.85	14.69	0.215	0.591	34.7
SCAD_tau_0.05	111.65	21.41	0.213	0.913	26.8
SCAD_tau_0.1	106.98	20.71	0.212	0.880	29.7

Table 3: Excess return statistical tests and Sharpe Ratio analysis.

Strategies	Right-tailed t-test	Wilcoxon signed-rank test	$\triangle Sharpe$
EWP	0.816	0.463	-0.029
MVP	0.979	0.997	-0.066
MVO	0.973	0.994	-0.062
SCAD_tau_0.01	0.293	0.114	0.017
SCAD_tau_0.05	0.044	0.023	0.055
SCAD_tau_0.1	0.046	0.039	0.054

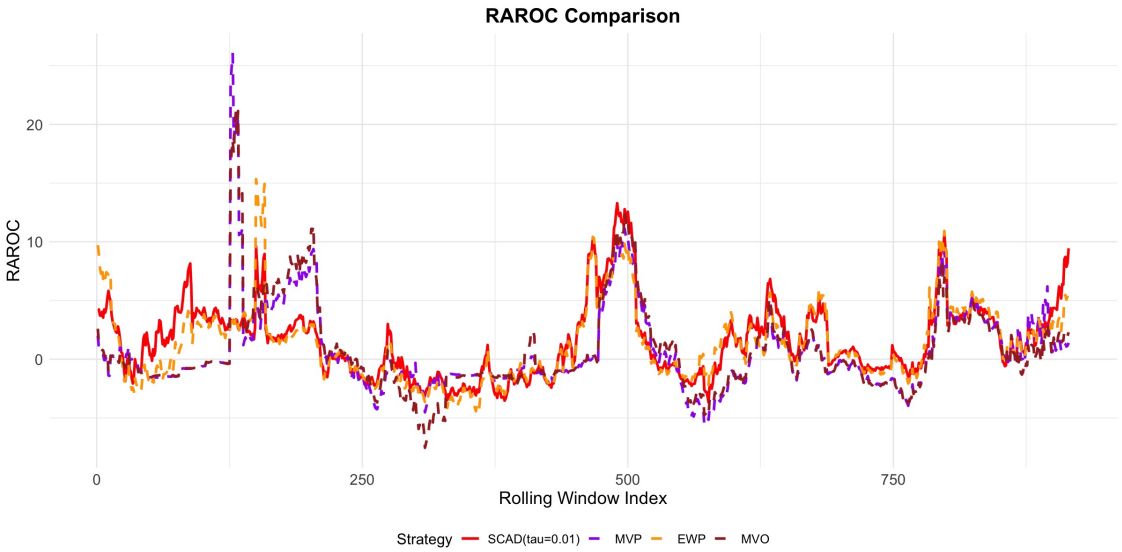
6.5. Risk-adjusted return analysis

Following [6, 37, 57, 61], we further evaluate performance using the Risk-Adjusted Return on Capital (RAROC) across different strategies. The RAROC is defined as the ratio of expected return to VaR at a given confidence level τ [61]:

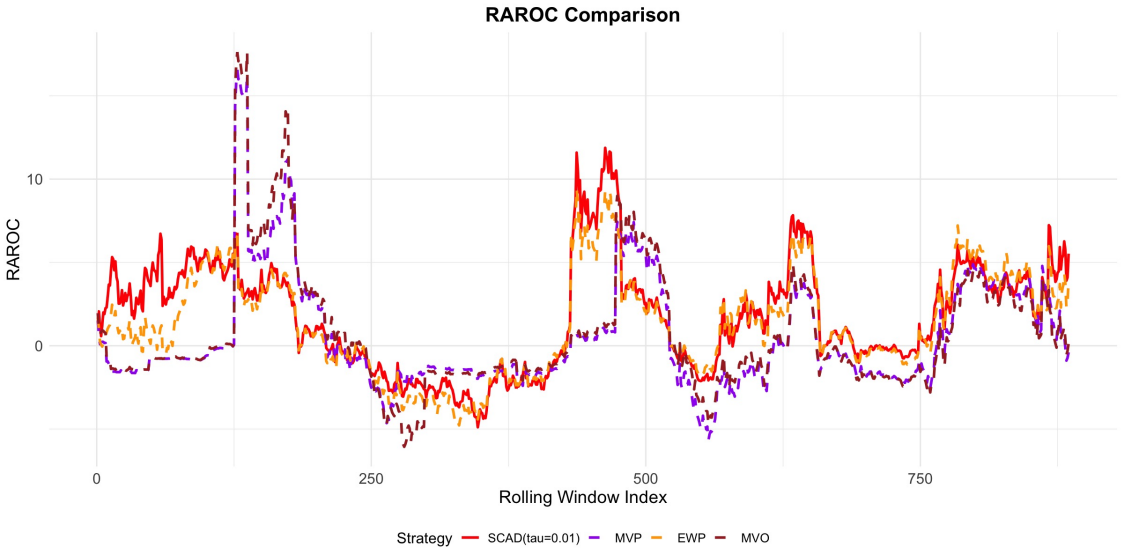
$$RAROC = \frac{\text{Expected Return}}{\text{VaR}_\tau}.$$

Originally introduced by Bankers Trust in the late 1970s, RAROC integrates profitability with the corresponding risk exposure, providing a standardized metric for comparing strategies. It has since been aligned with modern regulatory frameworks, including the Basel Committee's capital-adequacy standards (notably Basel III).

To assess the strategy's robustness across different investment horizons, we conduct a multi-period rolling backtest using 60-day and 90-day holding periods, each advanced by one trading day. Figure 6 compares the RAROC (99% confidence level) of our proposed framework against three standard benchmarks over these rolling windows. our proposed strategy consistently achieves higher RAROC values throughout most of the out-of-sample period. This outperformance holds across both the 60-day and 90-day horizons, underscoring the model's robustness and practical relevance for improving risk-adjusted returns. All strategies exhibit noticeable fluctuations in RAROC over time, reflecting shifts between bull, bear, and high-volatility regimes. These variations highlight differences in how each strategy adapts to changing market conditions. Notably, our proposed strategy not only limits losses during downturns but also captures market upswings effectively, accelerating return accumulation during recovery phases. As a further robustness check, Appendix C.4 reports the corresponding results at 90% and 95% confidence levels, which corroborate the core findings presented here.



(a) RAROC at 99% confidence level across portfolios: 60 holding days..



(b) RAROC at 99% confidence level across portfolios: 90 holding days.

Figure 6: RAROC at 99% confidence level across portfolios: 60 and 90 holding days.

Table 4 provides summary statistics of RAROC at different confidence levels (99%, 95%, 90%) for the 60-day and 90-day horizons. Our strategy exhibits highest mean and median RAROC and a highest proportion of positive outcomes ($> 62\%$) compared to traditional strategies, which often show negative median RAROC. Extending the holding period from 60 to 90 days amplifies the advantage of the SCAD framework. Besides, extending the holding period further amplifies the advantage of the SCAD framework: the 90% confidence RAROC rises 32% ($3.156 \rightarrow 4.164$) with only a 14.7% risk increase, demonstrating effectiveness for risk compensation. In contrast, MVP's mean RAROC deteriorates from 0.651 to 0.015 with horizon extension. These results challenge classical portfolio optimization efficacy and confirm our approach's advantages for long-term allocation.

Table 4: Summary statistics of RAROC across portfolios for 60-day and 90-day holding periods.

Confidence Interval	Strategy	60-day holding period				90-day holding period			
		Mean	Std	Median	Positive Ratio	Mean	Std	Median	Positive Ratio
99%	SCAD_tau_0.01	1.565	3.202	1.086	0.623	2.764	3.229	1.635	0.635
	EWP	1.274	3.231	0.768	0.586	1.298	2.998	0.930	0.616
	MVO	0.728	3.846	-0.274	0.443	0.655	3.690	-0.624	0.420
	MVP	0.691	3.677	-0.274	0.439	0.491	3.377	-0.776	0.385
95%	SCAD_tau_0.05	2.318	4.115	1.560	0.656	2.848	4.592	2.052	0.666
	EWP	1.652	4.195	1.024	0.586	1.787	4.196	1.340	0.616
	MVO	0.698	4.765	-0.616	0.443	0.441	5.053	-1.109	0.420
	MVP	0.687	4.790	-0.659	0.439	0.313	4.776	-1.490	0.385
90%	SCAD_tau_0.1	3.156	5.583	2.147	0.673	4.164	6.405	3.704	0.684
	EWP	2.080	5.170	1.230	0.586	2.425	5.612	1.628	0.616
	MVO	0.704	6.090	-0.897	0.443	0.205	6.968	-1.432	0.420
	MVP	0.651	6.129	-0.952	0.439	0.015	6.933	-2.128	0.385

Table 5 reports the corresponding statistics for the Sharpe ratio. Our strategy, particularly with $\tau = 0.1$, again delivers the highest mean and median Sharpe ratios (0.075/0.070 for 60-day; 0.068/0.074 for 90-day) and the highest frequency of positive ratios (above 68.9% and 71%, respectively). Conversely, traditional strategies underperform consistently. MVO and MVP yield negative mean and median, and positive ratios below 45% (MVP: 39.4% at 90 days). While extending to 90 days reduces volatility (standard deviation down 15%-20%) for all strategies, the performance ranking remains unchanged. This extension further increases SCAD's advantage in Sharpe ratios, and exacerbates underperformance of MVO and MVP, validating sparse optimization's effectiveness in balancing risk and return. These findings are fully consistent with Table 4.

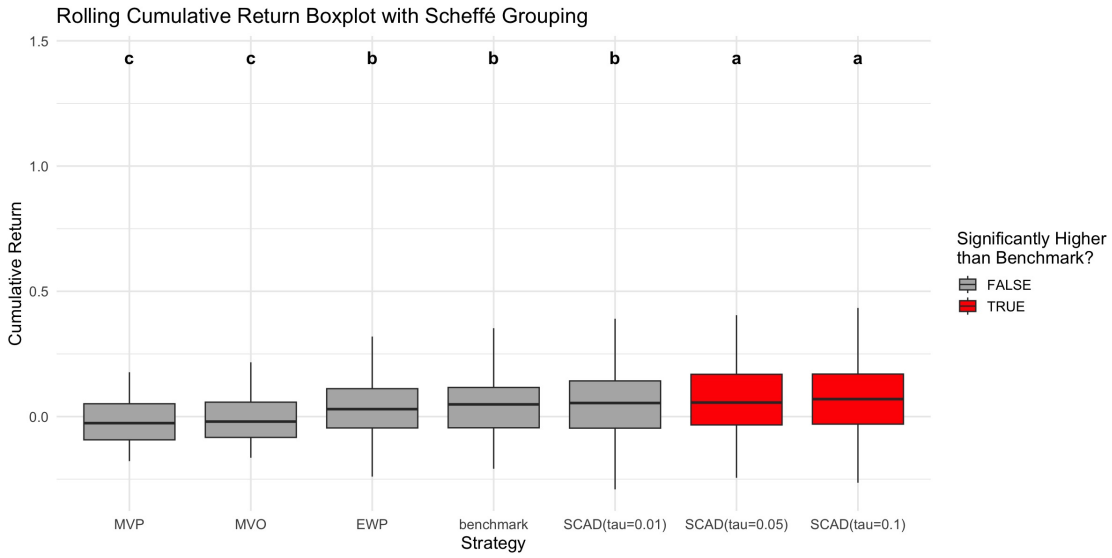
Table 5: Summary statistics of Sharpe ratio across portfolios for 60-day and 90-day holding periods.

Strategy	60-day holding period				90-day holding period			
	Mean	Std	Median	Positive Ratio	Mean	Std	Median	Positive Ratio
SCAD_tau_0.01	0.056	0.126	0.058	0.638	0.051	0.102	0.059	0.666
SCAD_tau_0.05	0.071	0.121	0.061	0.671	0.066	0.101	0.060	0.703
SCAD_tau_0.1	0.075	0.129	0.068	0.689	0.070	0.107	0.074	0.711
EWP	0.041	0.126	0.040	0.599	0.040	0.101	0.037	0.641
MVO	-0.010	0.139	-0.016	0.446	-0.012	0.112	-0.027	0.438
MVP	-0.013	0.134	-0.018	0.448	-0.017	0.108	-0.042	0.394

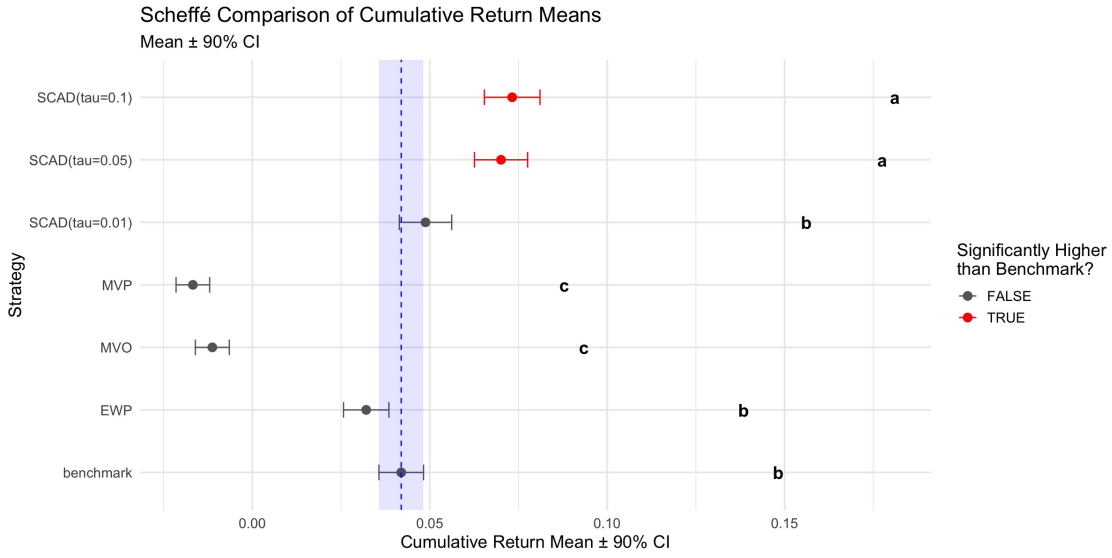
To statistically confirm the outperformance, we employ the Scheffé test for multiple comparisons against the CSI 300 benchmark for cumulative returns, RAROC, and Sharpe ratios. The test results in Figures 7–9 demonstrate that the proposed SCAD strategy delivers statistically superior cumulative returns, RAROC (90% confidence level), and Sharpe ratios relative to all benchmark portfolios over 90-day horizon. This statistical significance holds consistently across all comparisons. In contrast, the traditional strategies—EWP, MVP, and MVO—show no statistically significant advantage in any of the tests, with their performance consistently failing to exceed the relevant significance thresholds.

Thus, rigorous statistical testing validates that the SCAD-based strategy not only numerically outperforms the benchmarks (as shown in tables 4 and 5) but also achieves this outperformance with statistical significance, confirming its superior return and risk-adjusted performance. Appendix C.4 further reports the Scheffé test results for different holding pe-

riods. For the 60-day horizon, we present findings for RAROC at the 90%, 95%, and 99% confidence levels, cumulative return, and the Sharpe ratio. For the 90-day horizon, results are provided for RAROC at the 95% and 99% confidence levels. These findings further confirm the robustness and superior performance of our proposed strategy.

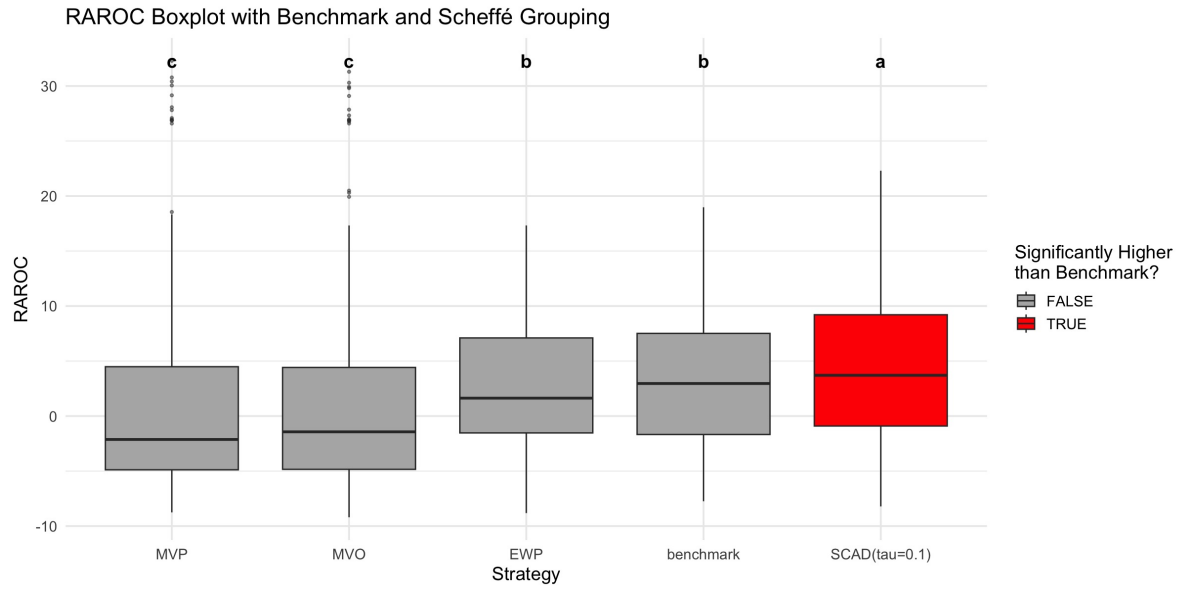


(a) Box plots of cumulative returns between portfolios.

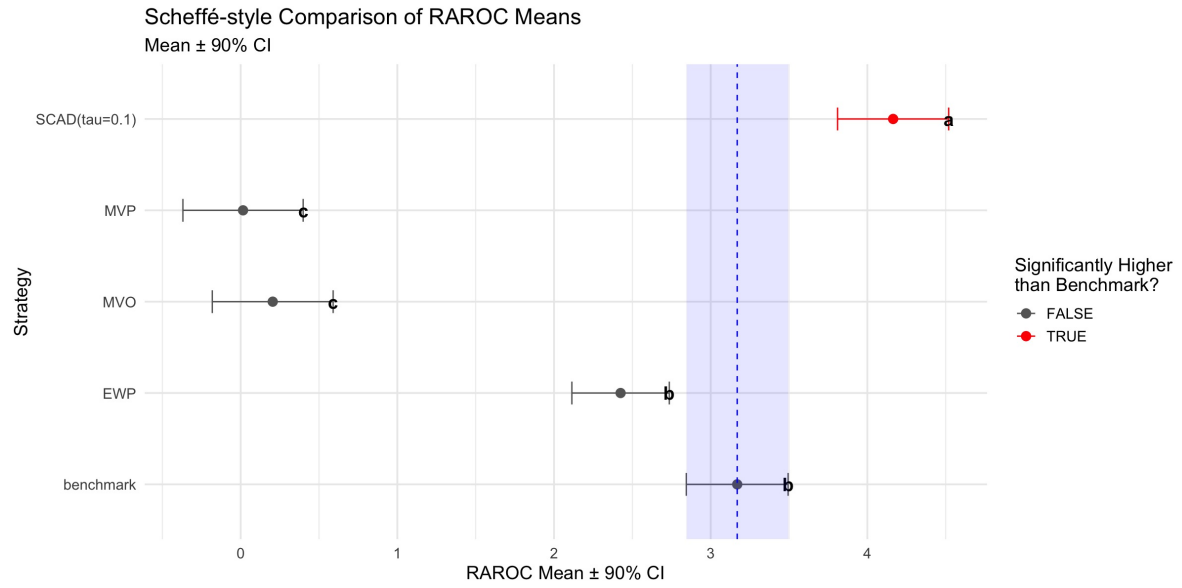


(b) Comparison of cumulative return significance between portfolios.

Figure 7: Scheffé test results for multiple comparisons of cumulative returns: 90 trading days.

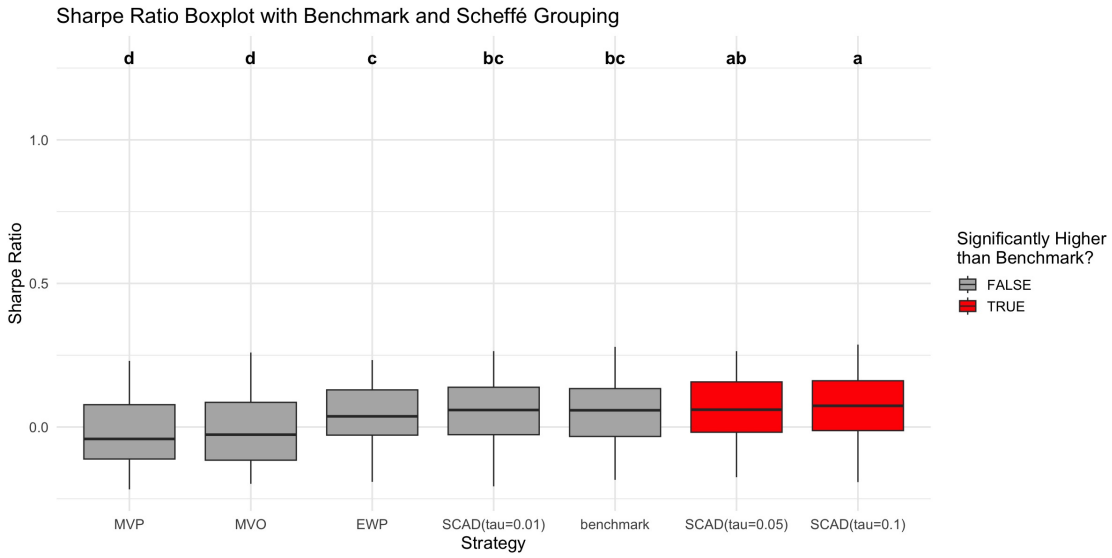


(a) Box plots of RAROC between portfolios.

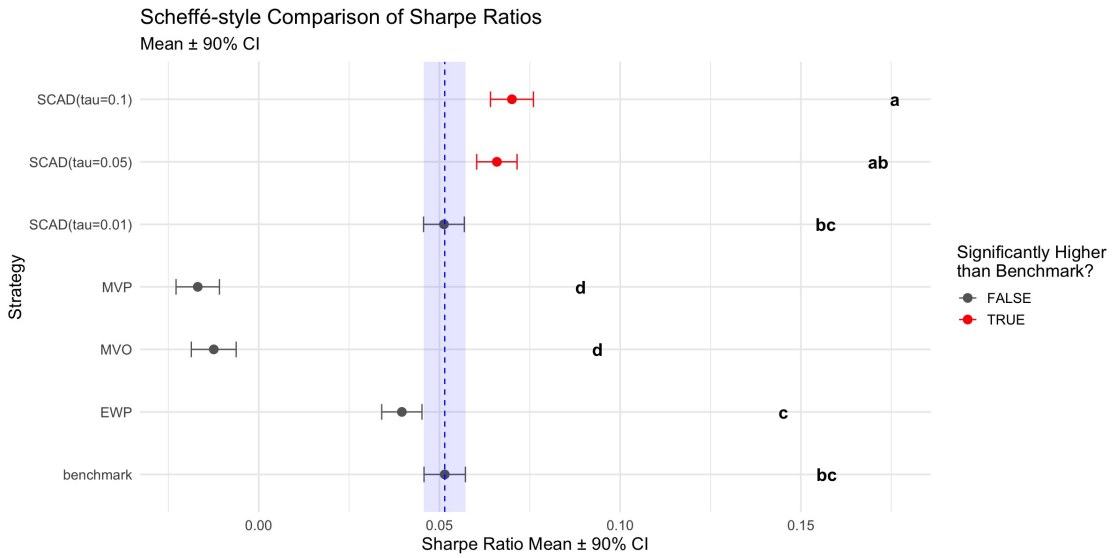


(b) Comparison of RAROC between portfolios.

Figure 8: Scheffé test results for multiple comparisons of RAROC at 90% confidence level: 90 trading days.



(a) Box plots of Sharpe ratio between portfolios.



(b) Comparison of Sharpe ratio significance between portfolios.

Figure 9: Scheffé test results for multiple comparisons of Sharpe ratio: 90 trading days.

6.6. Computational cost

We assess the computational cost of the proposed end-to-end framework, which is structured into three sequential blocks corresponding to the seven steps outlined in subsection 4.2:

- Part 1 (Steps 1-2): Logarithmic returns are calculated, and for each asset, an ARMA-GARCH model is estimated and combined with EVT to obtain the marginal distribution. This yields the uniformly transformed data:

$$\hat{u}_{it} = \hat{F}_i(\hat{z}_{it}), \quad \hat{u}_{it} \sim U(0,1), \quad i = 1, \ldots, p, \quad t = t_0 - L + 1, \ldots, t_0.$$

This transforms the original data into a uniformly distributed variable, facilitating subsequent modeling.

- Part 2 (Steps 3-6): Processes transformed data $\hat{u}_{it}$ using a 500-cycle rolling window and models the joint distribution with an R-vine copula, followed by 1000 simulations to forecast one-day-ahead returns r.
- Part 3 (Step 7): Constructs a high-dimensional portfolio optimization model using forecasted returns, applying adaptive-LASSO penalized quantile regression to determine optimal weights, and computes VaR and CVaR values.

Table 6: The computational cost of running the models.

	All parts (Steps 1-7)	Part 1 (Steps 1-2)	Part 2 (Steps 3-6)	Part 3 (Step 7)
The first trading day	494.79 seconds	343.60 seconds	108.02 seconds	43.17 seconds
The average of first 10 trading days	497.80 seconds	339.82 seconds	109.75 seconds	48.23 seconds
The average of first 50 trading days	498.68 seconds	341.02 seconds	110.62 seconds	47.04 seconds
The average of first 100 trading days	507.17 seconds	343.70 seconds	111.01 seconds	47.46 seconds

Table 6 details the execution times for its constituent parts, averaged over different points in the backtest. Part 1 (marginal modeling) is the most computationally intensive, requiring approximately 340 seconds, which is attributable to the sequential estimation of ARMA-GARCH-EVT models for each asset. Part 2 (copula fitting and simulation) completes in about 110 seconds, and Part 3 (portfolio optimization) in about 47 seconds. Given the high dimensionality of the problem and the complexity of the integrated models, this computational cost is practically feasible for daily portfolio rebalancing on standard hardware, confirming the operational viability of our framework.

7. Conclusions

This study employs an ARMA-GARCH model with residuals following a t-distribution to capture the dynamic mean and volatility of asset returns for one-day-ahead forecasting. We apply EVT to characterize the tail behavior of asset returns, and then implement an R-vine copula to model the interdependencies among multiple asset returns. By integrating a SCAD penalty, our approach provides a robust forecasting model capable of identifying optimal portfolio strategies even in high-dimensional settings. Furthermore, to address the non-convex optimization problem associated with such large-scale models, we utilize a penalized quantile regression algorithm to obtain solutions efficiently. Our proposed portfolio forecasting framework, which combines the R-vine copula with the SCAD penalty, is compared against traditional strategies such as the equal-weight portfolio, mean-variance optimization, and the minimum-variance portfolio.

Our empirical analysis uses two asset universes: ten sector indices from the SSE and 131 stocks from the CSI 300 Index, selected after filtering for data integrity, covering the period from January 4, 2010, to December 31, 2020. A rolling-window scheme with a 7:4 split divides the data into training and out-of-sample forecasting periods. To validate the R-vine copula's ability to capture multivariate dependence, we construct and visualize an

nine-layer tree structure for the ten SSE sectors. This structure clearly reveals the sector-level dependence network and potential risk transmission, confirming the model's flexibility in representing nonlinear and asymmetric dependencies in high-dimensional settings. Using the 131 CSI 300 stocks, we implement an integrated empirical framework that links stock return prediction to portfolio optimization.

To comprehensively evaluate the performance of the SCAD-based strategy, we conduct a multi-dimensional analysis covering portfolio sparsity, risk measures (VaR and CVaR), cumulative returns, risk-adjusted returns (RAROC and Sharpe ratio), and computational efficiency. The results show that the proposed strategy significantly enhances portfolio sparsity, effectively reduces risk, and delivers markedly higher cumulative returns. Its consistent outperformance relative to traditional benchmarks is confirmed through dynamic rolling metric plots, descriptive statistics, and Scheffé tests. Computational efficiency analysis further verifies the strategy's high operational speed and details its component-level complexity.

In summary, this multi-dimensional validation confirms that the proposed framework, which combines ARMA-GARCH, EVT, and R-vine copula modeling with SCAD-penalized sparse optimization, offers a practically effective and computationally tractable solution for high-dimensional portfolio management, balancing sparsity, risk control, and return enhancement.

Acknowledgments: Our optimization methodology extends the R package "rqPen" (Wu & Liu [69]), adapting its core components to meet our specific objectives. For R-vine copula estimation, we used "RVineStructureSelect," which implements the structure-selection algorithms developed by Brechmann et al. [14], Dissmann et. al. [22]. We would like to express our gratitude to the developers for their outstanding contributions. We also sincerely thank the Associate Editor and anonymous reviewers for their rigorous evaluations and insightful suggestions, which have significantly improved both the analytical rigor and clarity of this manuscript.

Declarations

Conflict of Interest Statement: The authors have no conflicts of interest related to the content of this article. No studies involving human participants or animals were conducted by any of the authors for this article.

Appendix A: Proof for Theorem 1.

Proof. From equation (16), the expected value of the quantile regression function is derived as follows:

$$\begin{aligned}
\mathbb{E}\left[\rho_\tau\left(Y-\xi\right)\right] &= \int_{-\infty}^{+\infty} \rho_\tau(y-\xi) dF_Y(y) \\
&= \int_{-\infty}^{+\infty} (y-\xi)[\tau - I(y<\xi)] dF_Y(y) \\
&= \tau \int_{-\infty}^{+\infty} (y-\xi) dF_Y(y) - \int_{-\infty}^{\xi} (y-\xi) dF_Y(y) \\
&= \tau(\mu-\xi) - \int_{-\infty}^{\xi} (y-\xi) dF_Y(y).
\end{aligned}$$

Here, we employ the fundamental properties $\mu = \mathbb{E}[Y]$ and $\int_{-\infty}^{+\infty} dF_Y(y) = 1$. These properties enable us to derive the final equation. Let us introduce the function $g(\xi)$ as follows:

$$g(\xi) := \tau(\mu-\xi) - \int_{-\infty}^{\xi} (y-\xi) dF_Y(y).$$

The calculation of the derivatives of $g(\xi)$ yields:

$$g'(\xi) = -\tau + F_Y(\xi), \quad g''(\xi) = f_Y(\xi) \geq 0,$$

which indicates that $g(\cdot)$ is convex due the no-negativity of $g''(\xi)$. Hence, the optimal value $\xi_\tau = F^{-1}(\tau)$ minimizes $g(\cdot)$, and

$$\min_\xi g(\xi) \implies \xi_\tau = \arg\min_\xi g(\xi).$$

Evaluating $\mathbb{E}\left[\rho_\tau(Y-\xi_\tau)\right]$, we have:

$$\mathbb{E}\left[\rho_\tau(Y-\xi_\tau)\right] = \tau\mu - \int_0^\tau F_Y^{-1}(t) dt.$$

Incorporating this result into equation (11), we confirm that $\mathrm{CVaR}_\tau = \frac{1}{\tau}\mathbb{E}\left[\rho_\tau(Y-\xi_\tau)\right] - \mu$. □

Appendix B: EVT and Copula dependency structure analysis

Table 7 presents descriptive statistics for the daily log returns of the ten SSE sector indices; mean returns are shown in percent. The Energy sector shows the lowest average return (-0.042%), while Consumer Staples achieves the highest (0.055%). All series exhibit negative skewness (left-skewed distributions), and most kurtosis are greater than 3, indicating heavier tails than a normal distribution. The Jarque-Bera tests (significant at the 1% level) for all sectors confirm that returns deviate significantly from normality.

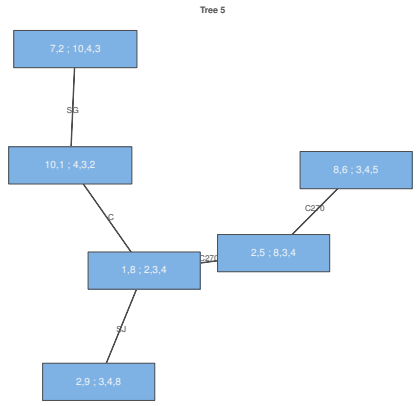
Table 7: Descriptive statistics for the daily log returns of sector indexes.

Series	Mean	max	min	std_{dev}	Skewness	Kurtosis	Range	JB
Energy	-0.042	0.087	-0.105	0.017	-0.466	4.420	0.192	2277***
Materials	-0.002	0.080	-0.096	0.018	-0.624	4.118	0.176	2067***
Industrials	0.003	0.082	-0.095	0.016	-0.560	5.497	0.177	3533***
Cons Disc	0.013	0.068	-0.097	0.016	-0.753	4.121	0.165	2149***
Cons Staples	0.055	0.067	-0.090	0.016	-0.622	3.198	0.157	1314***
Health Care	0.042	0.074	-0.096	0.017	-0.526	3.142	0.170	1225***
Financial	0.011	0.087	-0.103	0.016	-0.104	5.035	0.190	2835***
Info Technology	0.024	0.074	-0.102	0.021	-0.471	1.945	0.176	521***
Communication Services	0.006	0.077	-0.105	0.020	-0.656	3.344	0.182	1441***
Utilities	-0.004	0.070	-0.088	0.014	-0.778	6.769	0.158	5383***

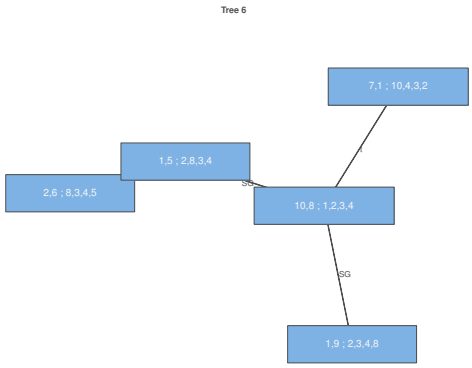
The total number of observations is 2674 for each stock index. The mean is expressed as a percentage. JB shows the result of Jarque-Bera's normality test.
***Denotes statistical significance at the 1% level.

Table 8: GARCH(1,1) parameters for daily log returns of sector indexes.

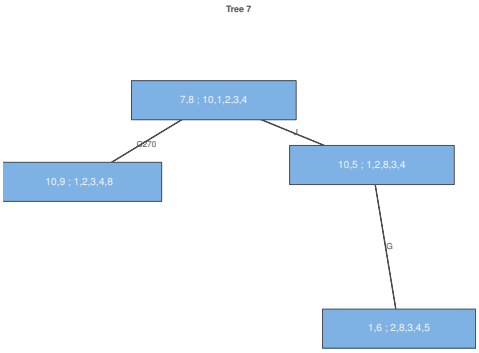
Series	μ	ω	α	β	λ
Energy	-1.55e-04	3.10e-06	0.056	0.933	1.177
Materials	3.32e-04	3.96e-06	0.065	0.922	1.328
Industrials	2.20e-05	2.96e-06	0.064	0.924	1.245
Cons Disc	4.31e-04	1.15e-06	0.066	0.932	1.365
Cons Staples	7.94e-04	3.78e-06	0.072	0.913	1.481
Health Care	6.20e-04	8.50e-07	0.060	0.939	1.521
Financial	-2.32e-04	2.06e-06	0.063	0.930	1.086
Info Technology	3.67e-04	2.33e-06	0.050	0.946	1.438
Communication Services	5.58e-04	5.46e-06	0.062	0.923	1.360
Utilities	-1.33e-06	1.03e-06	0.066	0.930	1.210



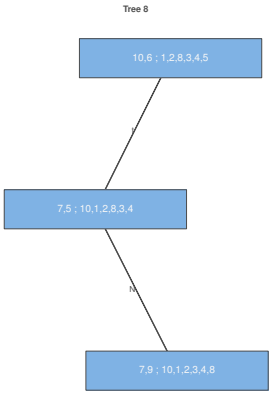
(a) The fifth tree in R-vine copula



(b) The sixth tree in R-vine copula.



(c) The seventh tree in R-vine copula.



(d) The eighth tree in R-vine copula

Figure 10: An illustative representation of R-vine copulas structures Tree 5–Tree 8.

Table 9 reports the estimated thresholds (μ), shape (ξ), and scale (β) parameters for the upper and lower tails of each sector's residual series. The results confirm the presence of heavy tails in the distributions across all sectors. A larger scale parameter β suggests greater dispersion of extreme observations from the threshold, indicating higher volatility in the tails.

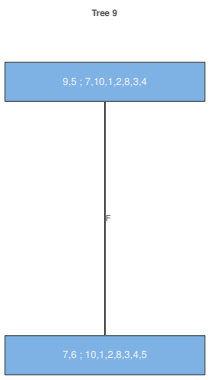


Figure 11: An illustative representation of R-vine copulas structures Tree 9

Table 9: EVT parameters for standardized residuals from ARMA(1,1)-GARCH(1,1)

Series	ARMA(1,1)-GARCH(1,1)					
	Upper tail			Lower tail		
	μ^U	ξ^U	β^U	μ^L	ξ^L	β^L
Energy	1.035	0.026	0.630	-1.488	-0.044	0.820
Materials	1.485	-0.044	0.558	-1.305	0.002	0.717
Industrials	1.462	-0.026	0.602	-1.897	0.011	0.889
Cons Disc	1.620	-0.029	0.465	-1.647	-0.009	0.791
Cons Staples	1.947	-0.185	0.441	-1.989	-0.093	0.804
Health Care	1.349	-0.078	0.529	-1.997	-0.069	0.697
Financial	0.957	-0.030	0.774	-1.093	0.007	0.675
Info Technology	1.373	-0.140	0.503	-1.362	-0.089	0.767
Communication Services	1.544	-0.004	0.485	-1.619	-0.088	0.810
Utilities	1.525	0.100	0.483	-2.061	0.070	0.920

Appendix C: Supplementary experimental results

C.1. Supplementary results on sparsity analysis

We now discuss the use of generalized cross-validation (GCV) to estimate the optimal SCAD parameters $\theta = (\lambda, \alpha)$, as proposed in Fan & Li [26]. To reduce data dependence, we apply industry-based dimensionality reduction. Specifically, we group stocks into ten non-overlapping sectors based on the CSI industry classification and 10 sector return series using asset weights. Due to significant heterogeneity across sectors—such as differences in policy environments, supply-demand structures, and technology cycles—10 sector returns can reasonably be treated as independent, satisfying the core assumptions of GSV.

The expression for the GCV criterion is given by:

$$\beta_{\mathrm{GCV}}(\theta) = \left(Z^T Z + n\Sigma_\lambda(\beta_0)\right)^{-1} Z^T y,$$

where Z is the matrix composed of the return series of the ten sectors, with each column representing one sector's return series. Subsequently, one can calculate the fitted values $\hat{y} = Z\beta_{\mathrm{GCV}}(\theta)$, and define the projection matrix $P_Z(\hat{\beta}(\theta))$ as:

$$P_Z(\hat{\beta}(\theta)) = Z\left(Z^T Z + n\Sigma_\lambda(\beta_0)\right)^{-1} Z^T.$$

We determine the number of effective parameters $e(\theta)$ in the context of the penalized least squares fit as the trace of the projection matrix:

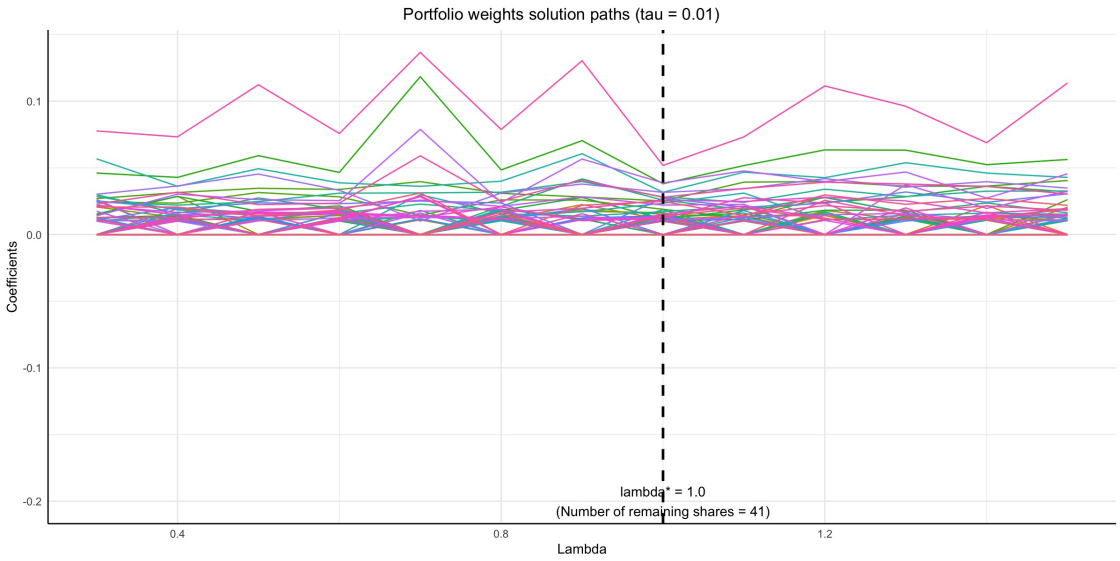
$$e(\theta) = \mathrm{tr}\left[P_Z(\hat{\beta}(\theta))\right].$$

Consequently, we compute the GCV statistic as follows:

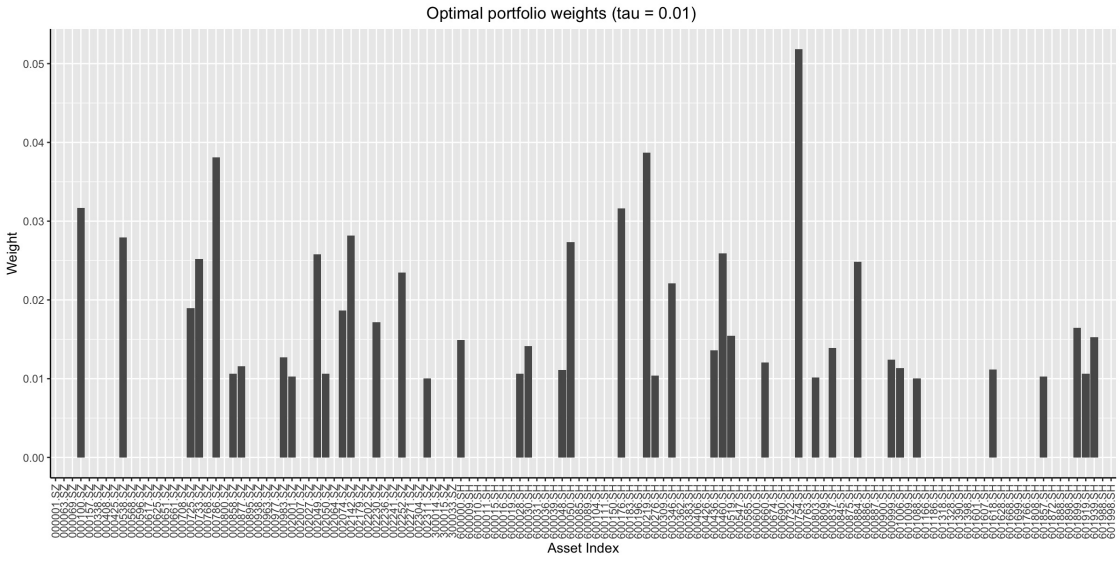
$$GCV(\theta) = \frac{1}{n}\frac{\|y - Z\hat{\beta}(\theta)\|^2}{\left(1 - \frac{e(\theta)}{n}\right)^2},$$

where the optimal parameters $\hat{\theta}$ are those that minimize the GCV function

$$\hat{\theta} = \arg\min_{\theta} GCV(\theta).$$

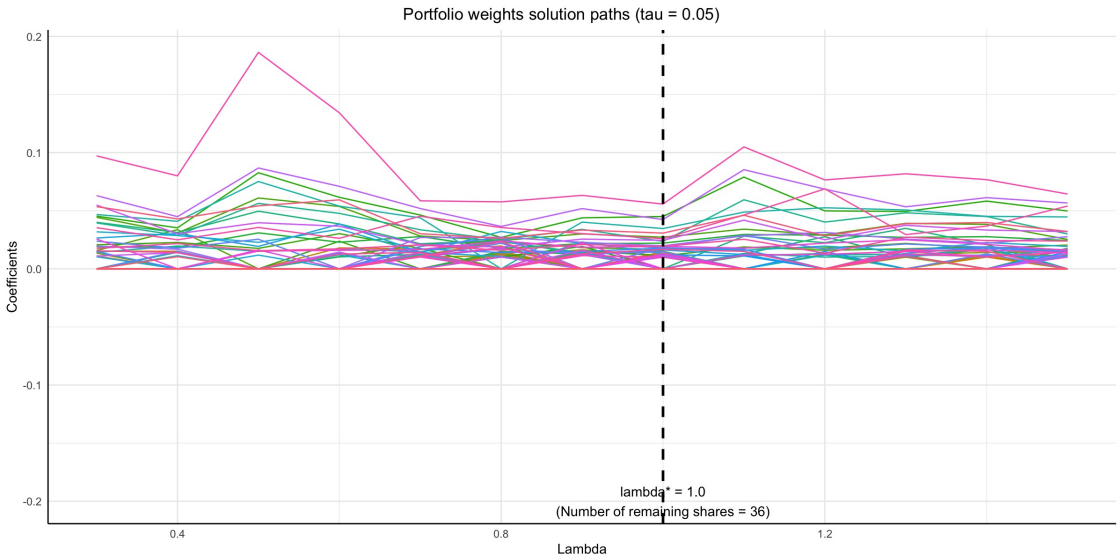


(a) Portfolio weight paths at $\tau = 0.01$.

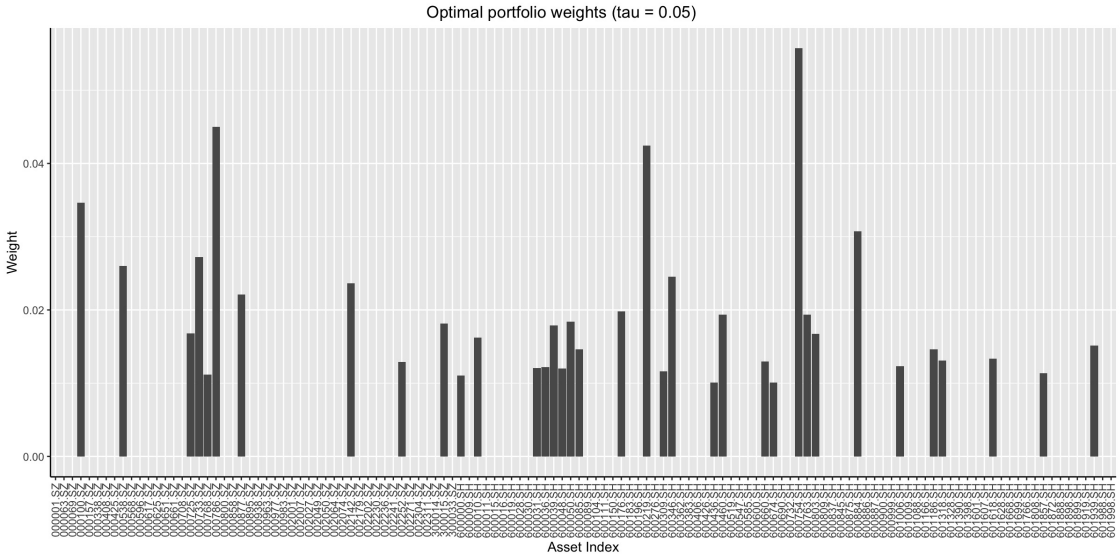


(b) Portfolio weights at $\lambda = 1$ and $\tau = 0.01$.

Figure 12: Portfolio weight paths at $\tau = 0.01$ and weights at $\lambda = 1$.

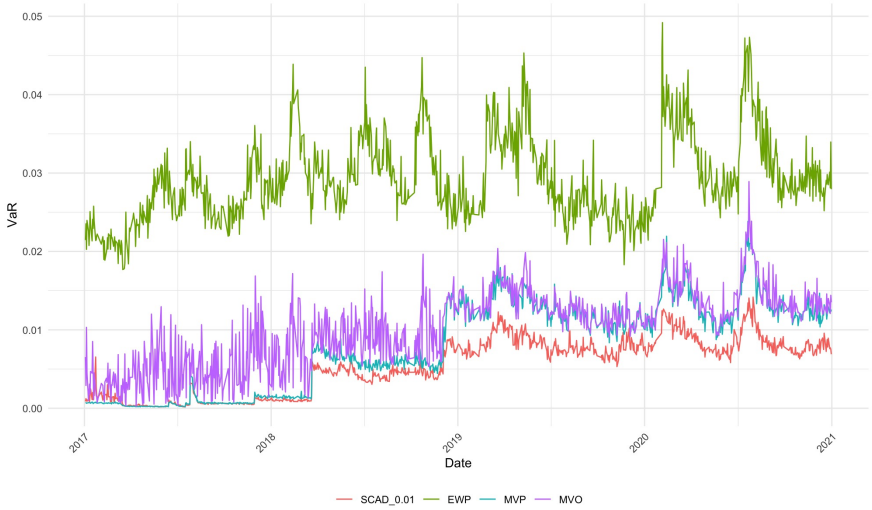


(a) Portfolio weight paths at $\tau = 0.05$.

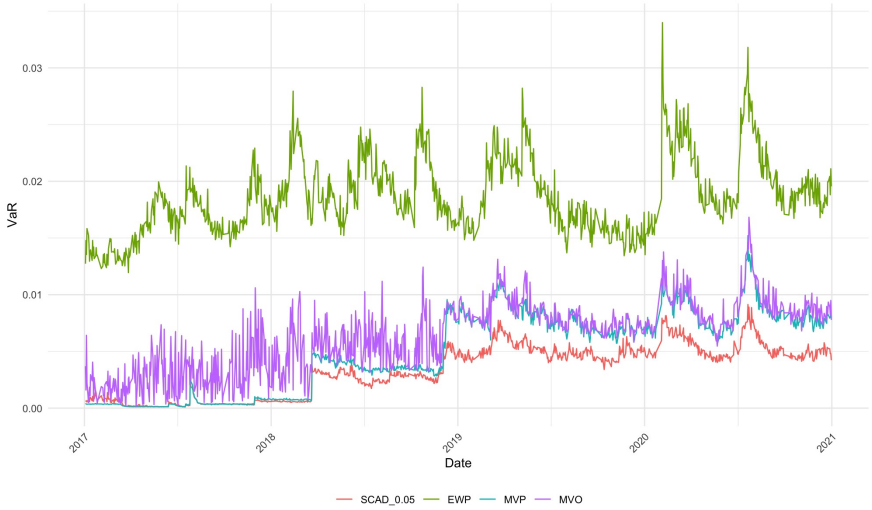


(b) Portfolio weights at $\lambda = 1$ and $\tau = 0.05$.

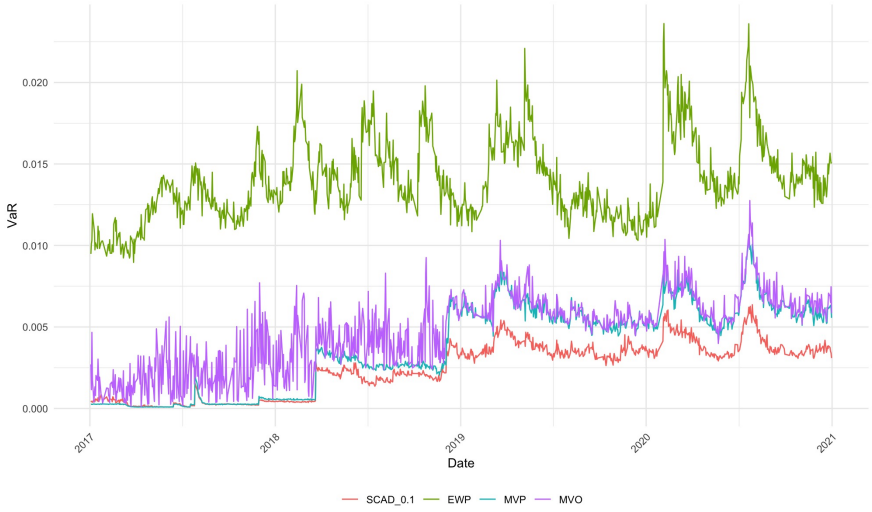
Figure 13: Portfolio weight paths at $\tau = 0.05$ and weights at $\lambda = 1$.



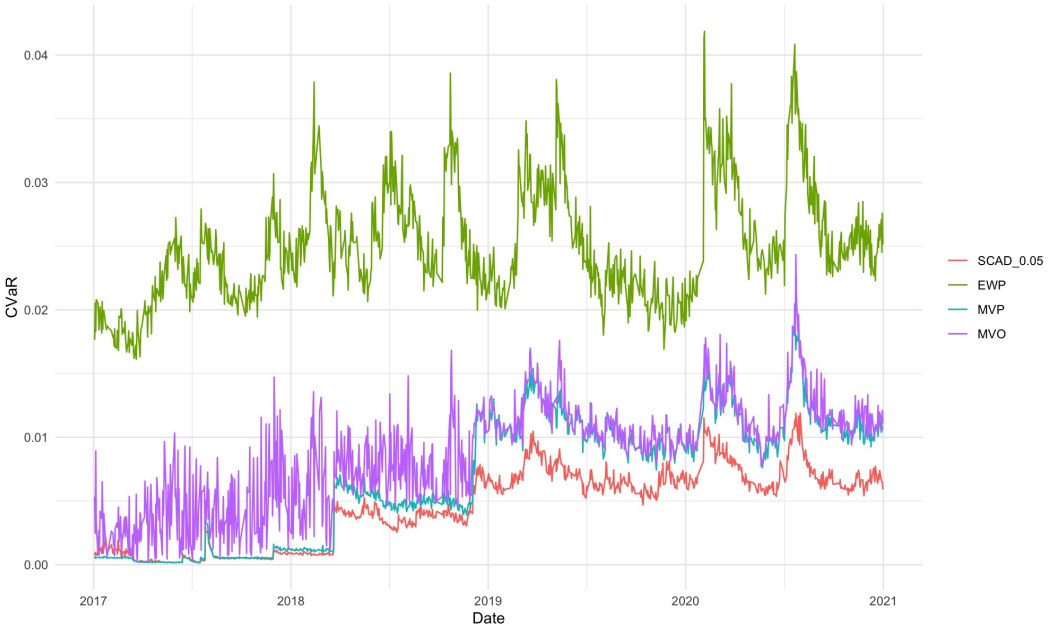
(a) VaR at 99% confidence level.



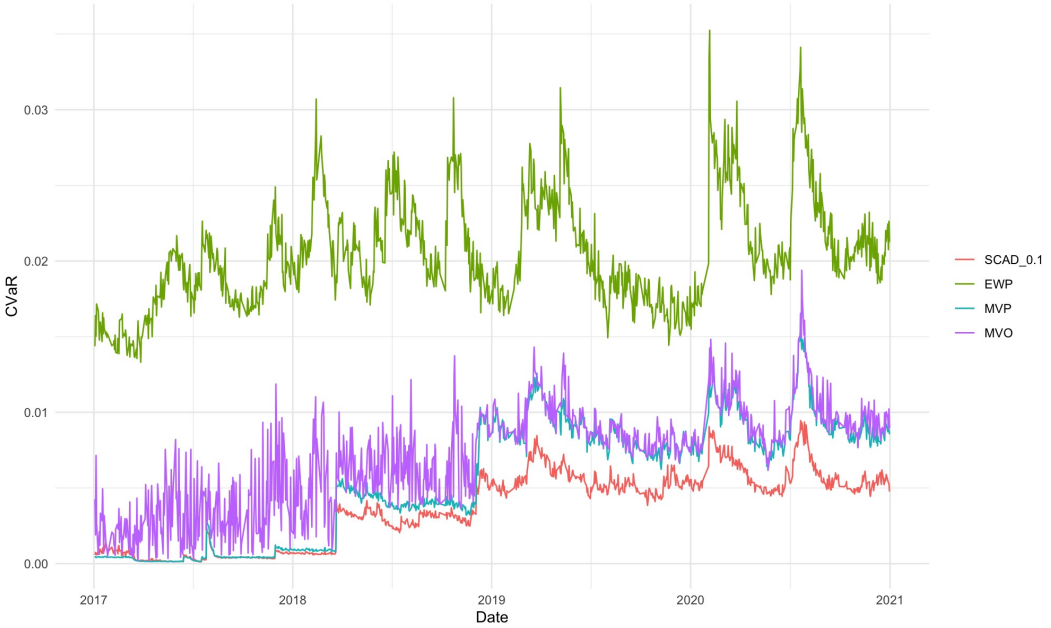
(b) VaR at 95% confidence level.



(c) VaR at 90% confidence level.

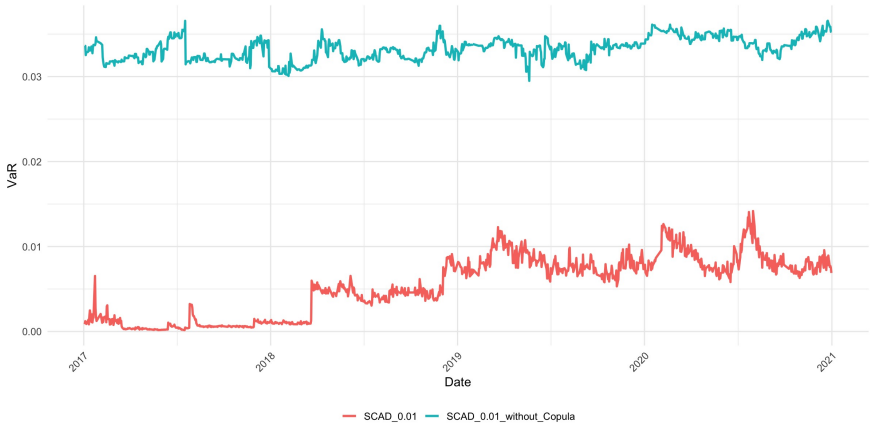


(a) CVaR at 95% confidence level.

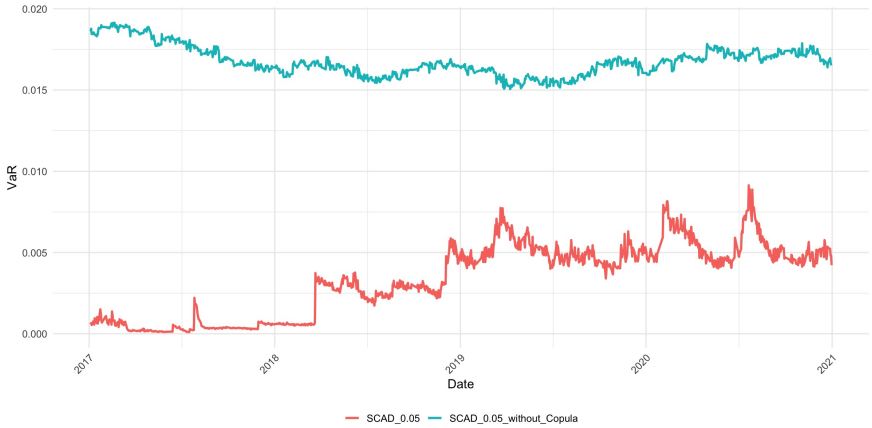


(b) CVaR at 90% confidence level.

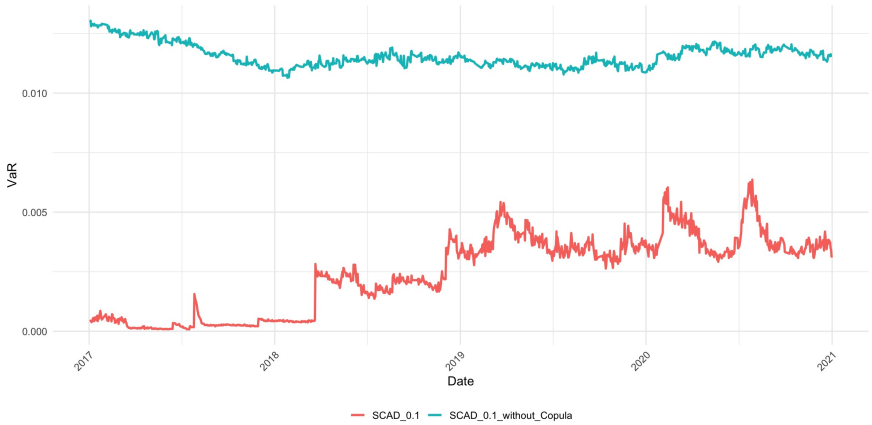
Figure 15: Comparison of CVaR across portfolios.



(a) Forecasted VaR at 99% confidence level.

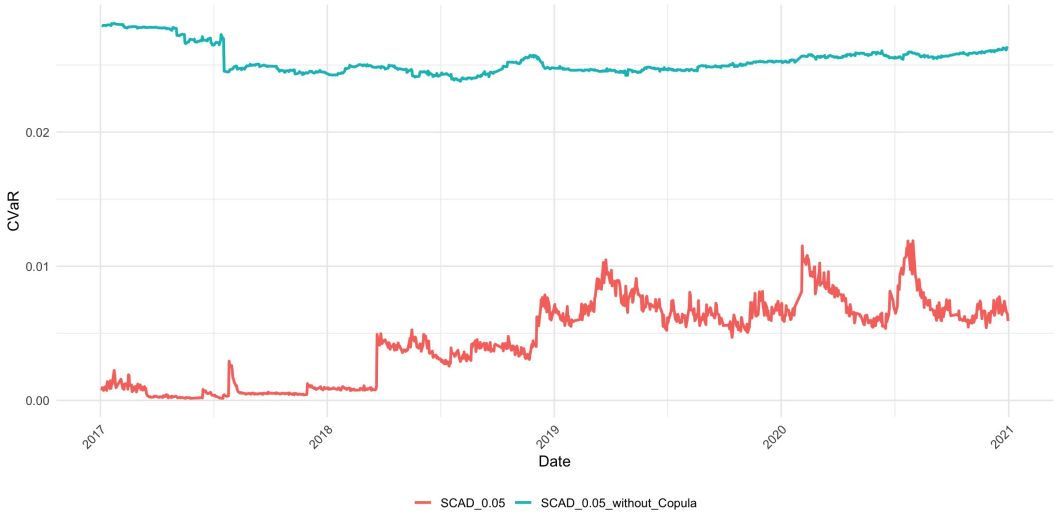


(b) Forecasted VaR at 95% confidence level.

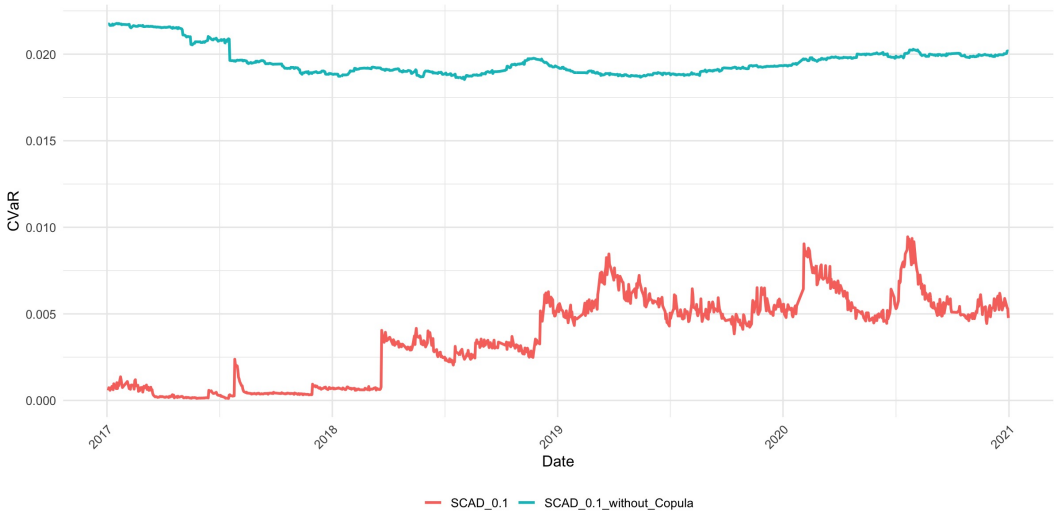


(c) Forecasted VaR at 90% confidence level.

Figure 16: Forecasted VaR: with vs. without R-vine copula.



(a) Forecasted CVaR at 95% confidence level.



(b) Forecasted CVaR at 90% confidence level.

Figure 17: Forecasted CVaR: with vs. without R-vine copula.

C.3. Supplementary results on cumulative returns over the 60-day and 90-day holding periods

Figures 18 and 19 exhibit the cumulative returns for the investment strategies over the 60-day and 90-day holding periods, respectively. The results indicate that the proposed SCAD strategy consistently generated higher cumulative returns compared to the traditional portfolio strategies (EWP, MVP, MVO) throughout the majority of the evaluation period. Furthermore, the SCAD strategy demonstrated well-suited capture of upside market movements, leading to higher overall returns. Notably, the performance advantage of the SCAD strategy increases with longer investment horizons.

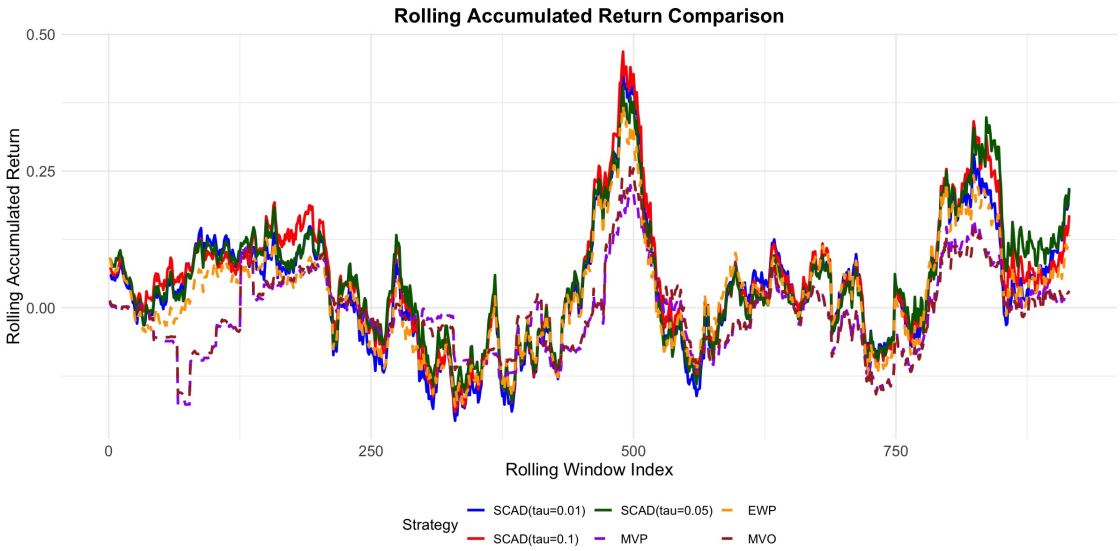


Figure 18: Cumulative return across portfolios: 60-day holding period.

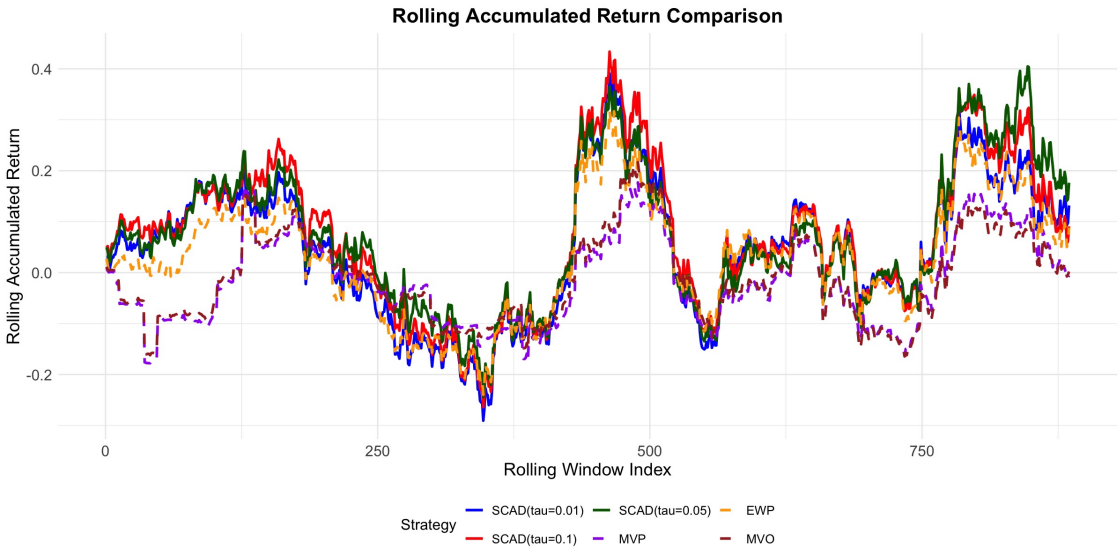
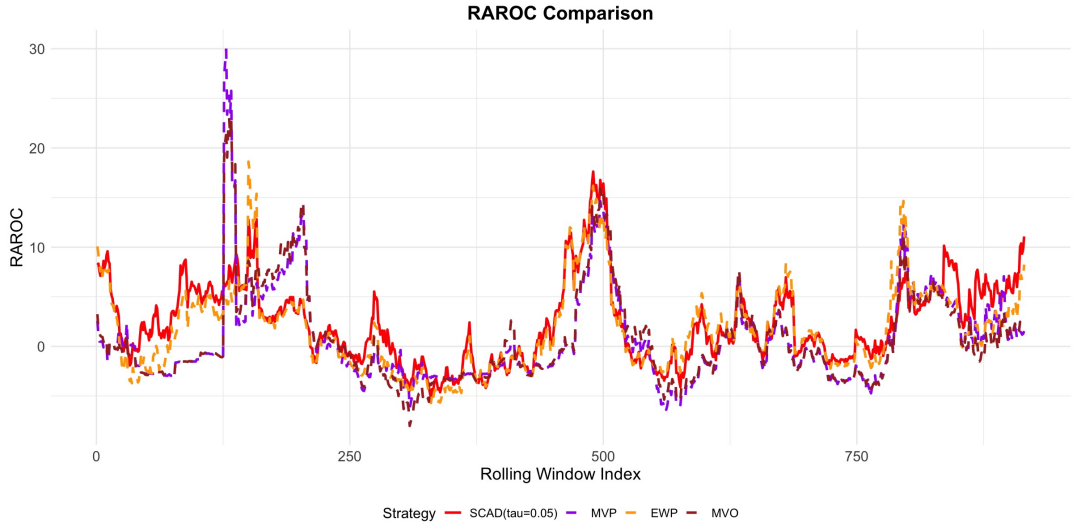
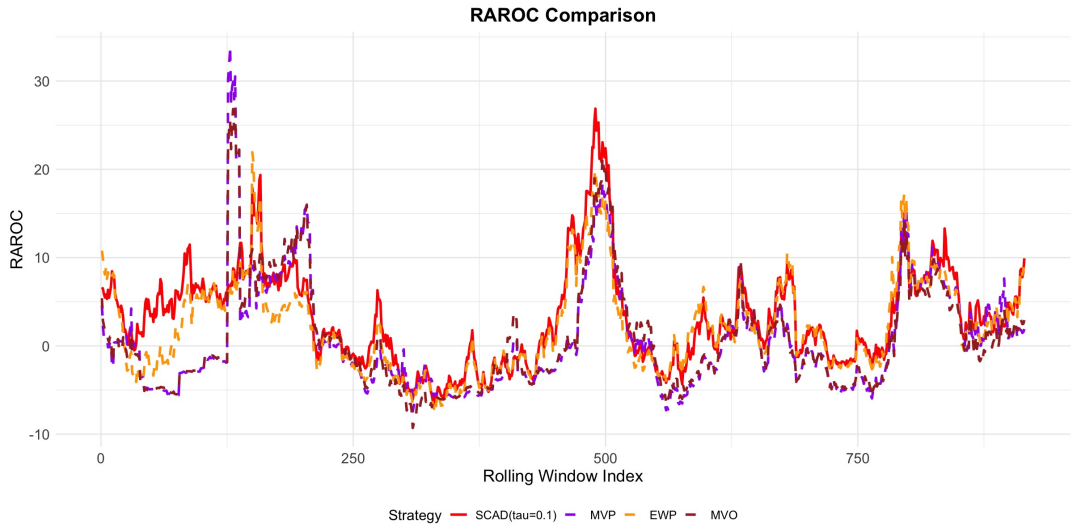


Figure 19: Cumulative return across portfolios: 90-day holding period.

C.4. Supplementary results on risk-adjusted return on capital analysis

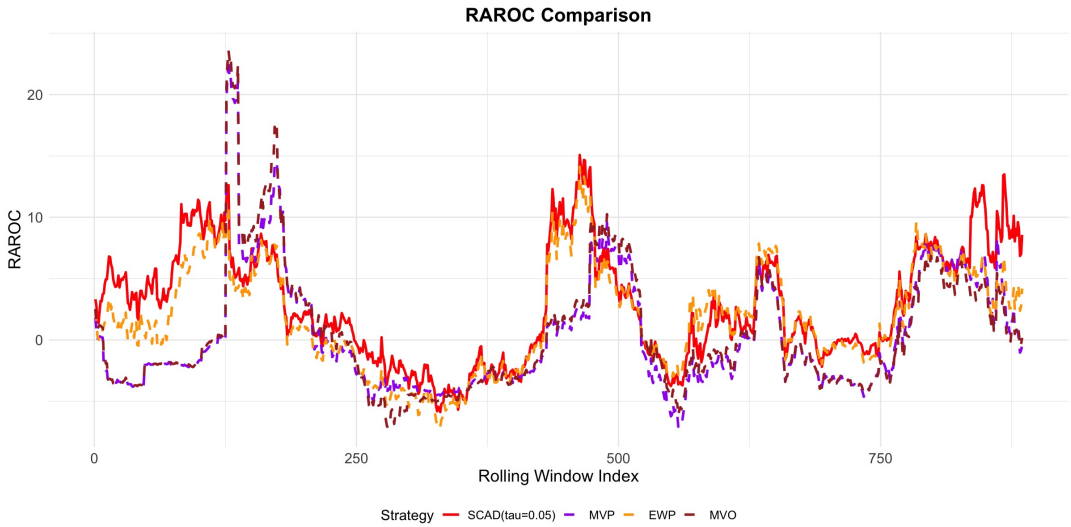


(a) RAROC at 95% confidence level.

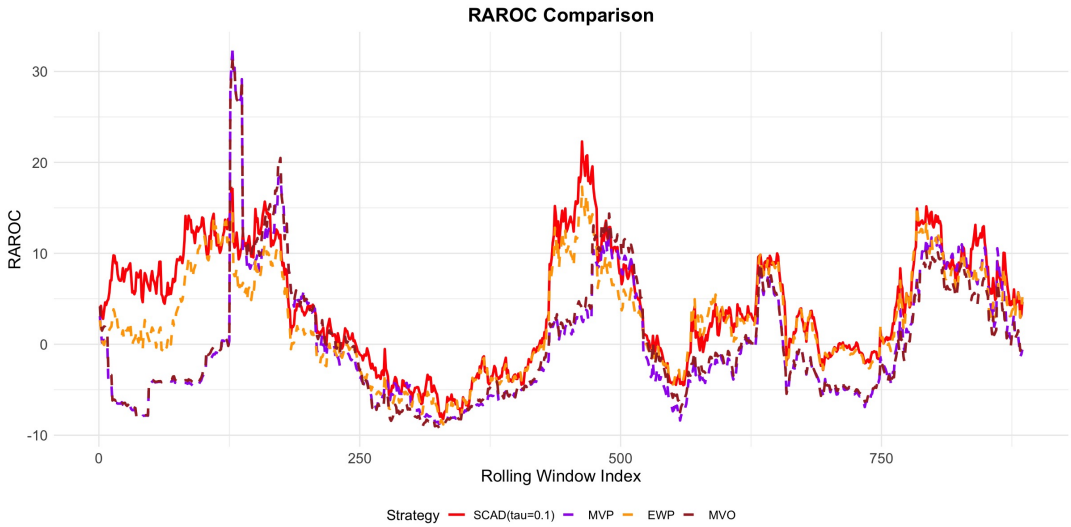


(b) RAROC at 90% confidence level.

Figure 20: RAROC at varying confidence levels across portfolios: 60 holding days.

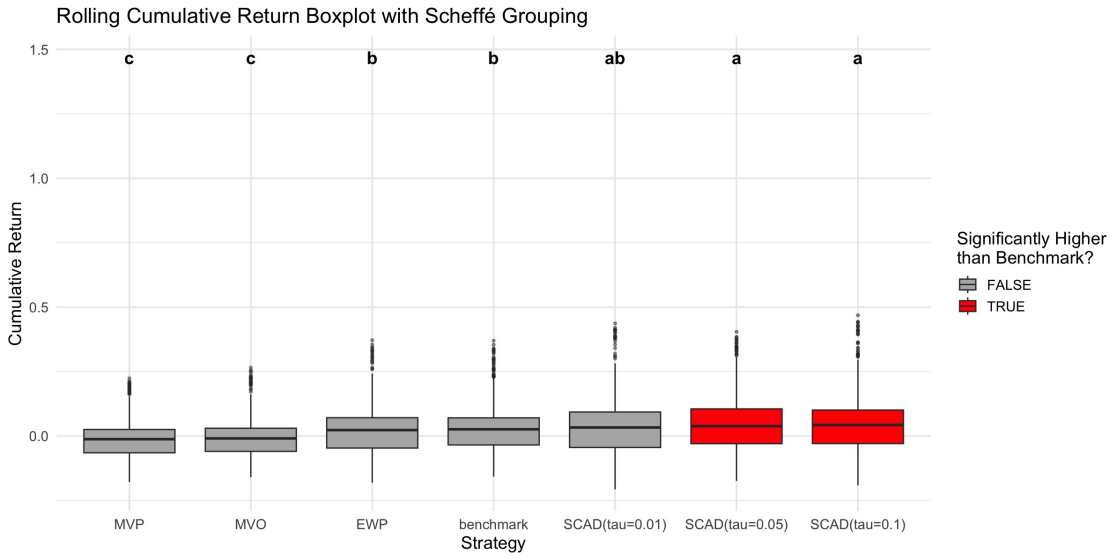


(a) RAROC under 95% confidence level.

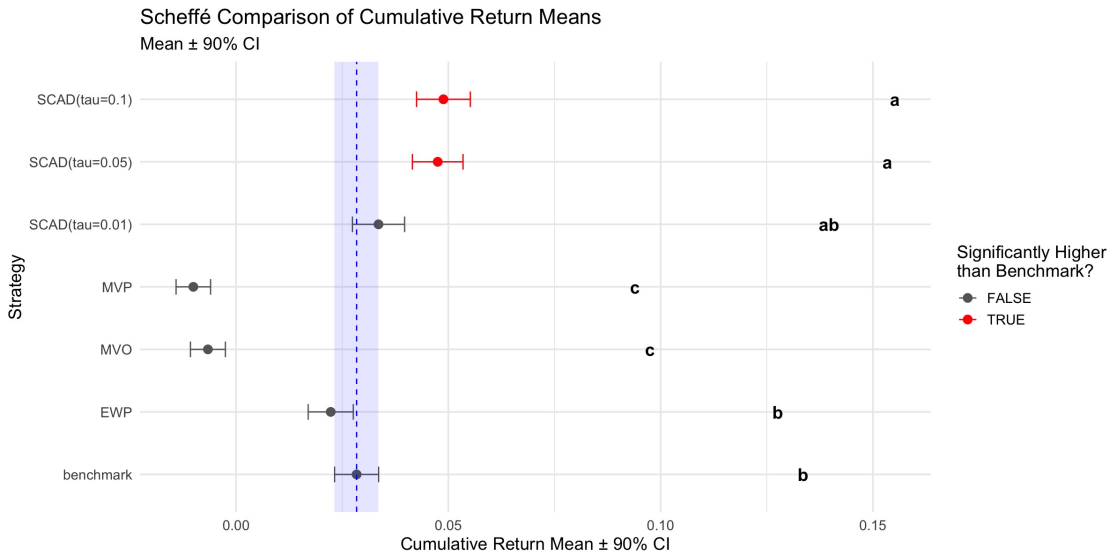


(b) RAROC under 90% confidence level.

Figure 21: RAROC at varying confidence levels across portfolios: 90 holding days.

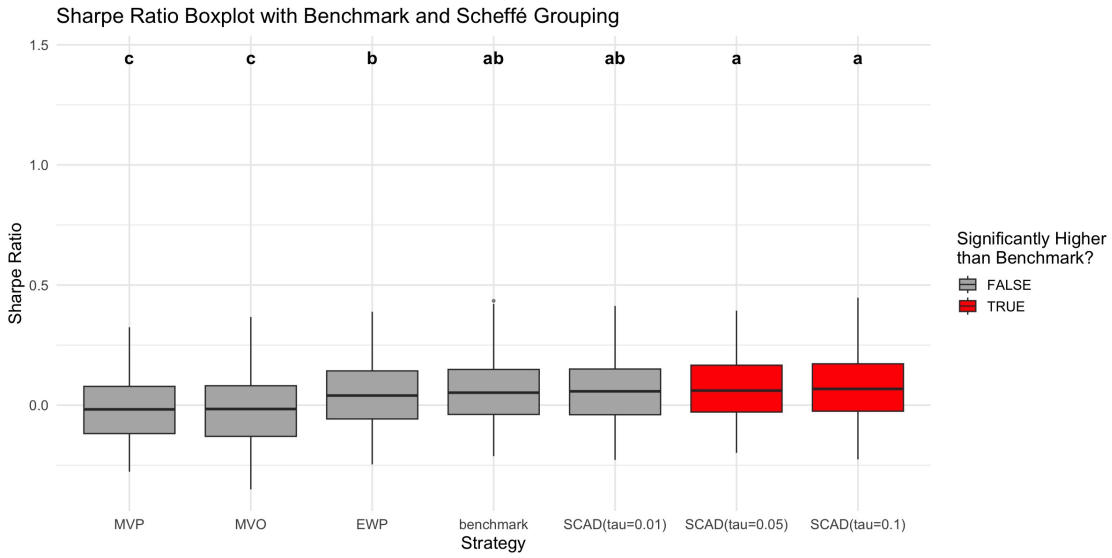


(a) Box plots of cumulative returns between portfolios.

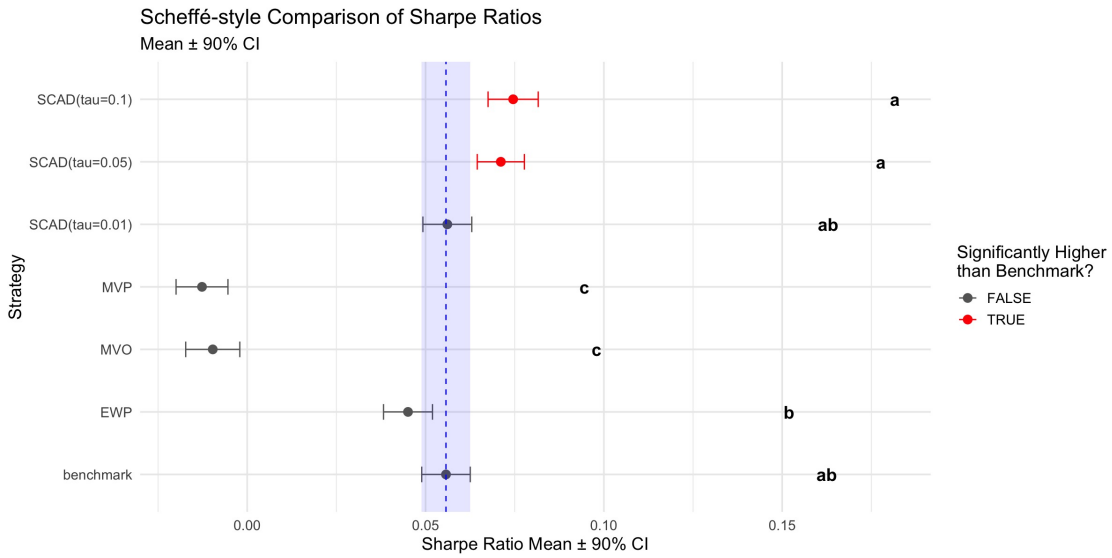


(b) Comparison of cumulative return significance between portfolios.

Figure 22: Scheffé test results of cumulative returns: 60 trading days.

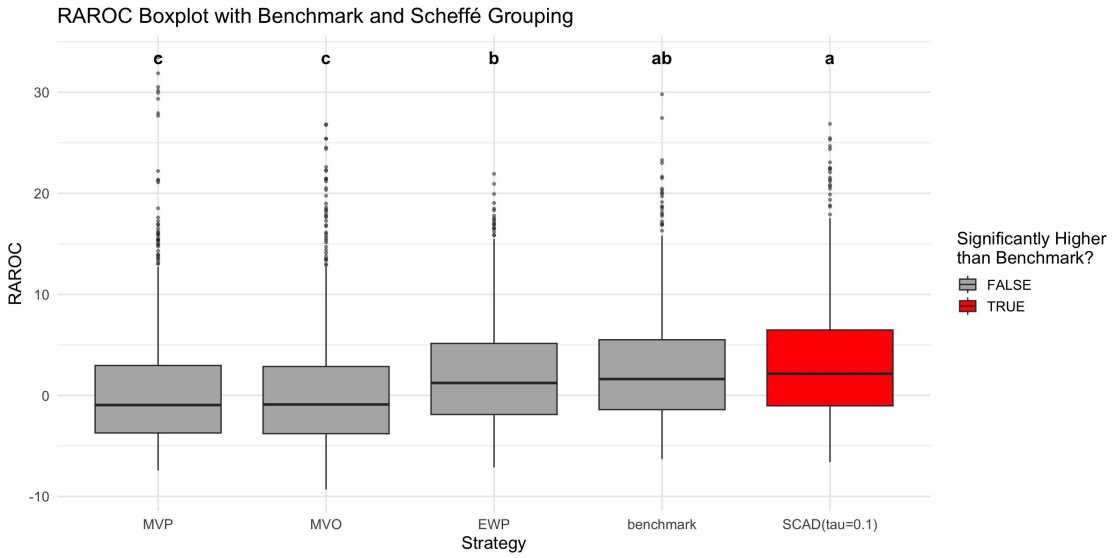


(a) Box plots of Sharpe ratio between portfolios.

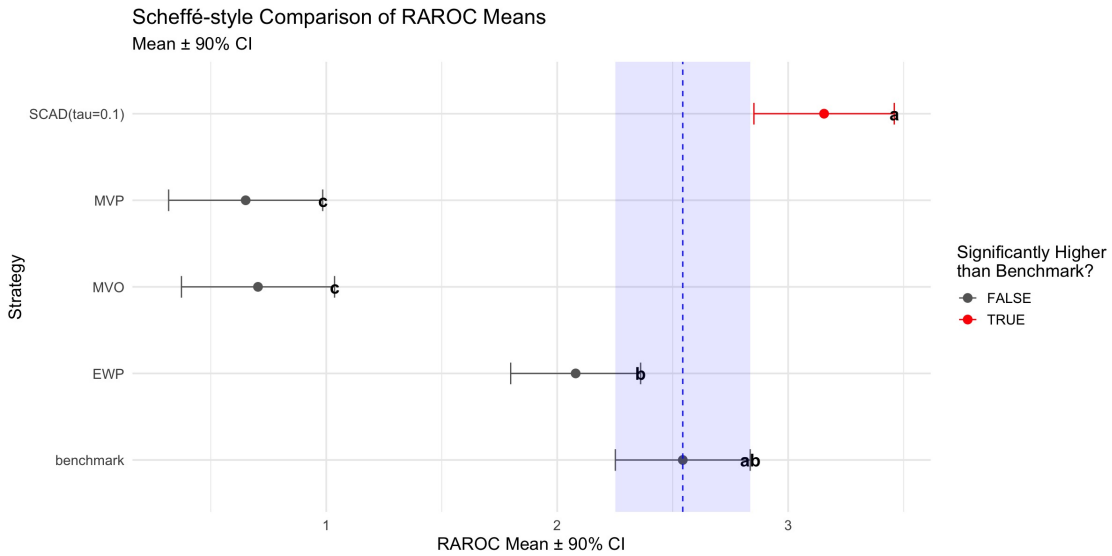


(b) Comparison of Sharpe ratio significance between portfolios.

Figure 23: Scheffé test results for multiple comparisons of Sharpe ratio: 60 trading days.

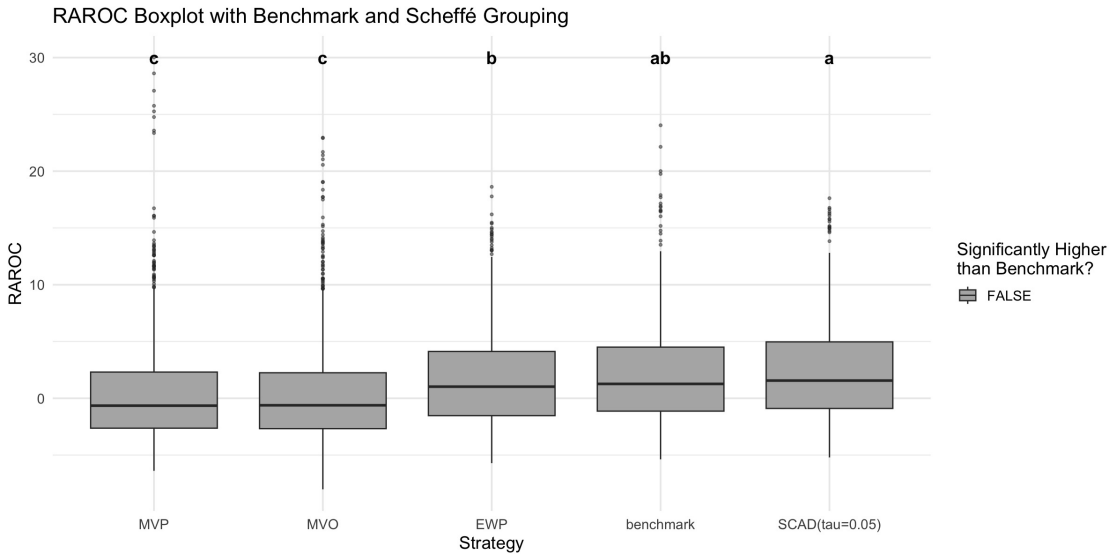


(a) Box plots of RAROC between portfolios.

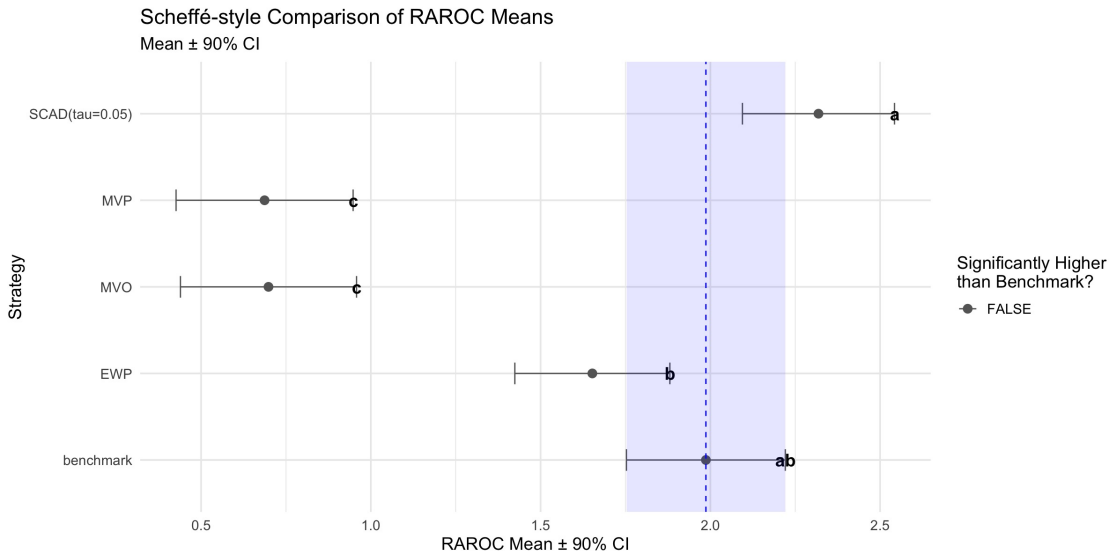


(b) Comparison of RAROC between portfolios.

Figure 24: Scheffé test results of RAROC at 90% confidence level: 60 trading days.

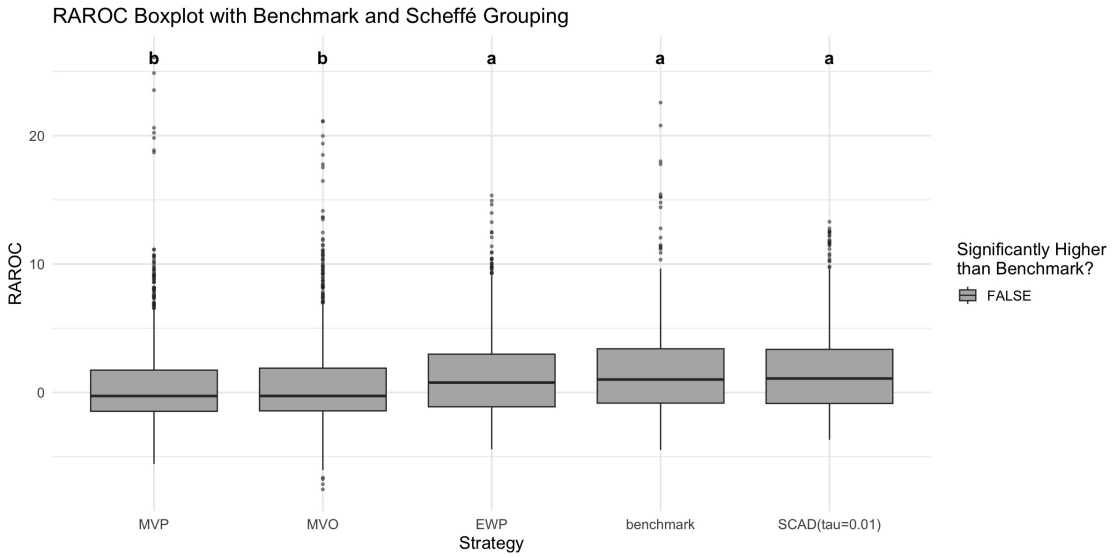


(a) Box plots of RAROC between portfolios.

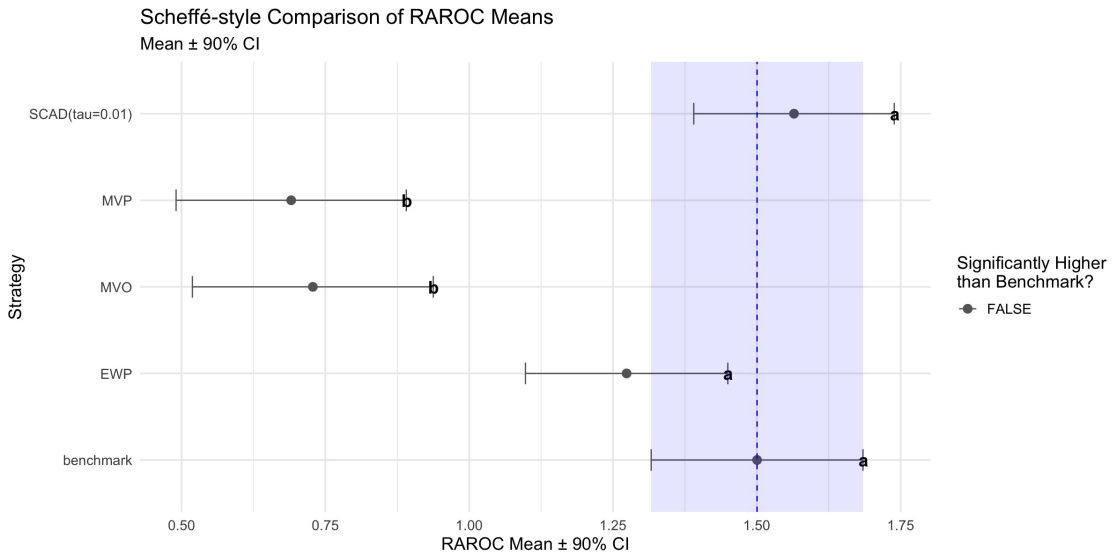


(b) Comparison of RAROC between portfolios.

Figure 25: Scheffé test results of RAROC at 95% confidence level: 60 trading days.

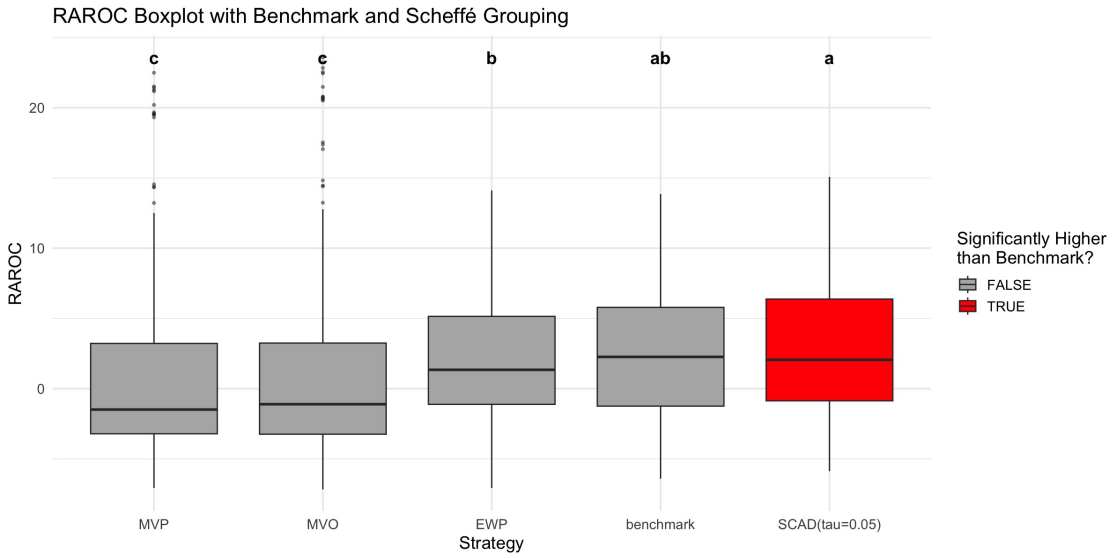


(a) Box plots of RAROC between portfolios.

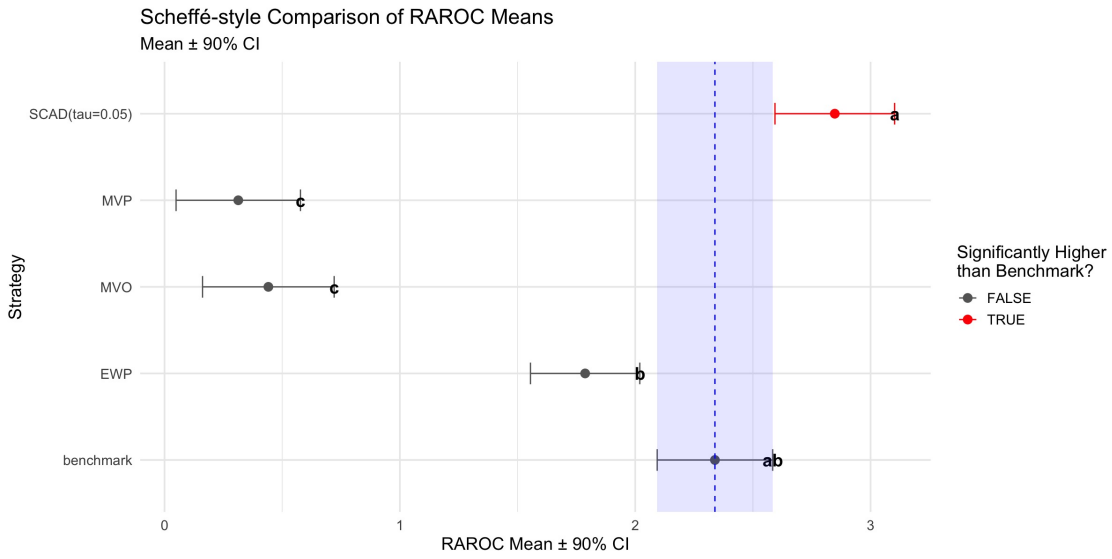


(b) Comparison of RAROC between portfolios.

Figure 26: Scheffé test results of RAROC at 99% confidence level: 60 trading days.

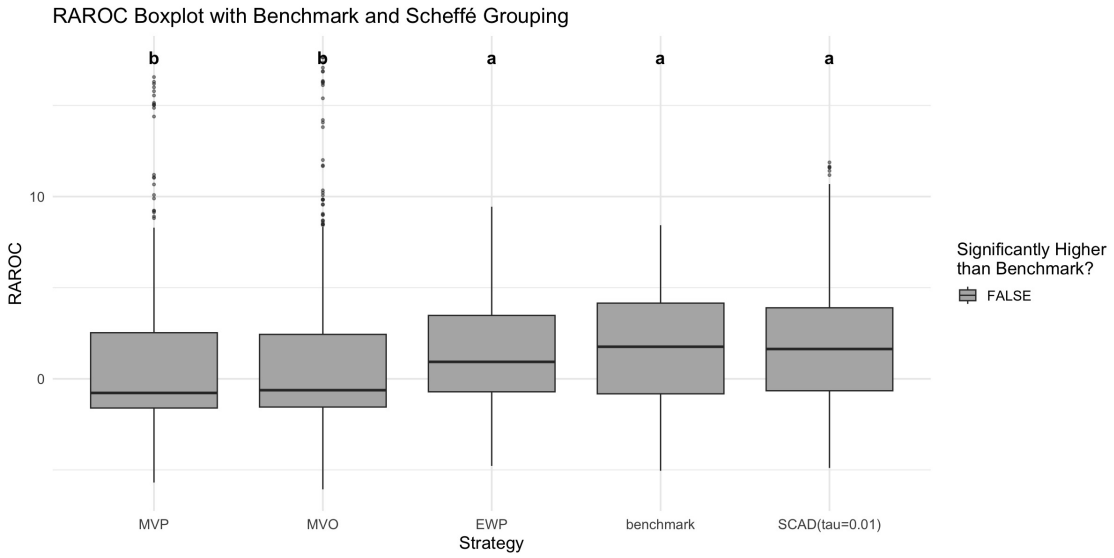


(a) Box plots of RAROC between portfolios.

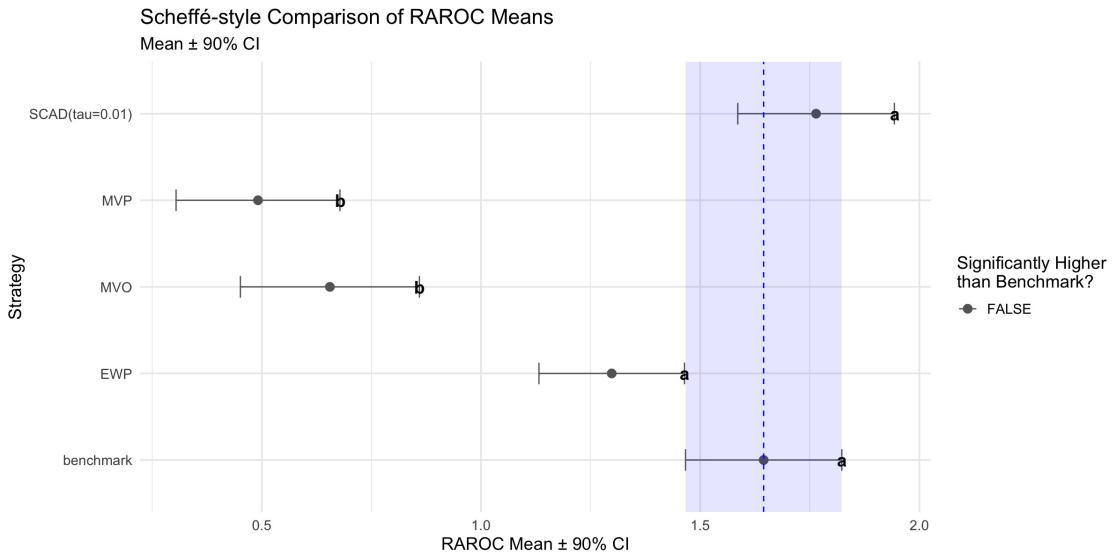


(b) Comparison of RAROC between portfolios.

Figure 27: Scheffé test results of RAROC at 95% confidence level: 90 trading days.



(a) Box plots of RAROC between portfolios.



(b) Comparison of RAROC between portfolios.

Figure 28: Scheffé test results of RAROC at 99% confidence level: 90 trading days.

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